

Lecture 21: Linear Regression I & II

Heather Mattie, PhD

November 13, 2025

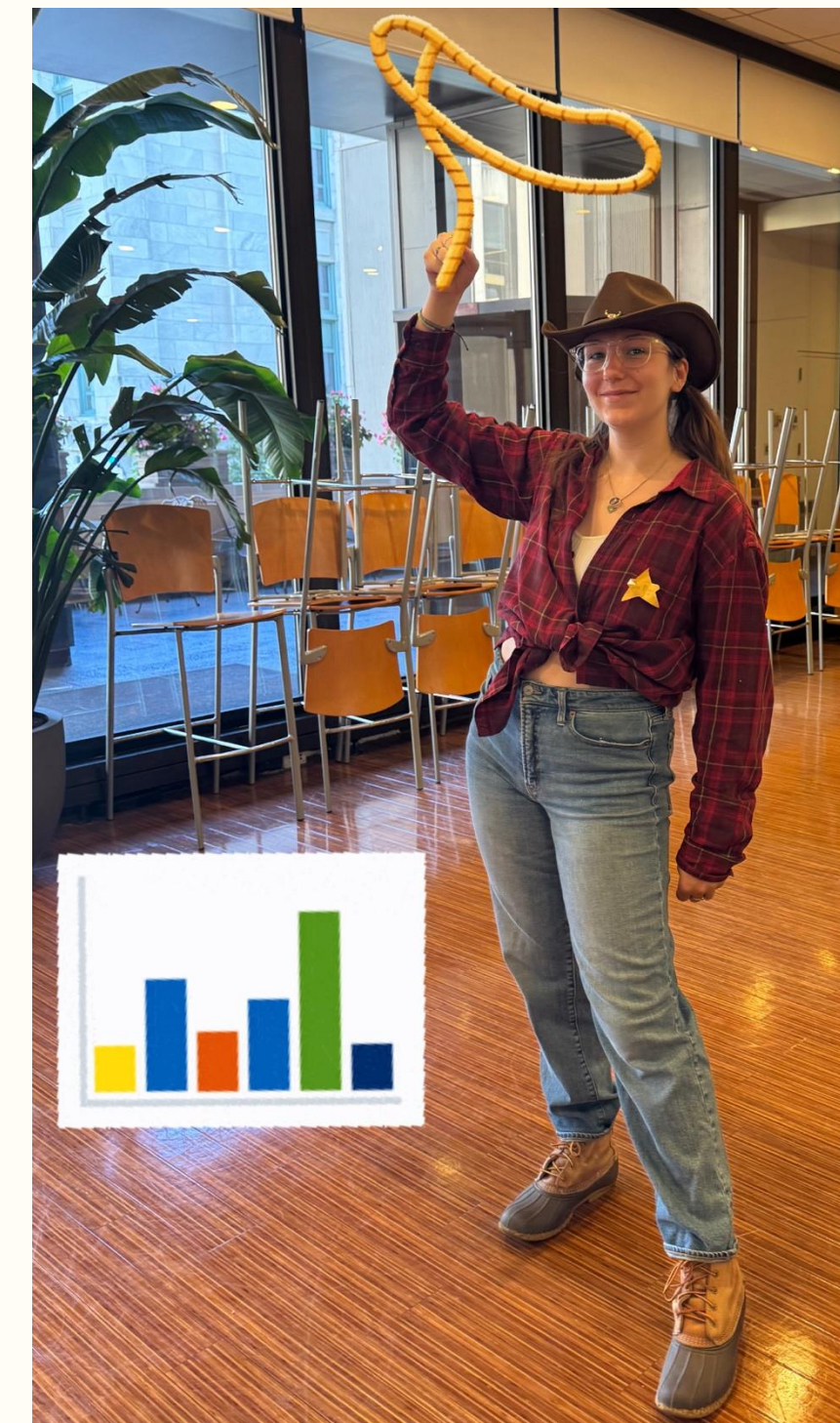


Recipe of the Day!

Cinnamon Rolls



Biostats Halloween Fun: **Ranger Dee Plyer**, the best data wrangler in Boston. “No dataset is too tough for a cowgirl like me to handle!”



Announcements

- No lab this week
- Project preference form due Friday, November 21st
- Homework #4 due Friday, December 5th
- No class November 25 and 27

Linear Regression I

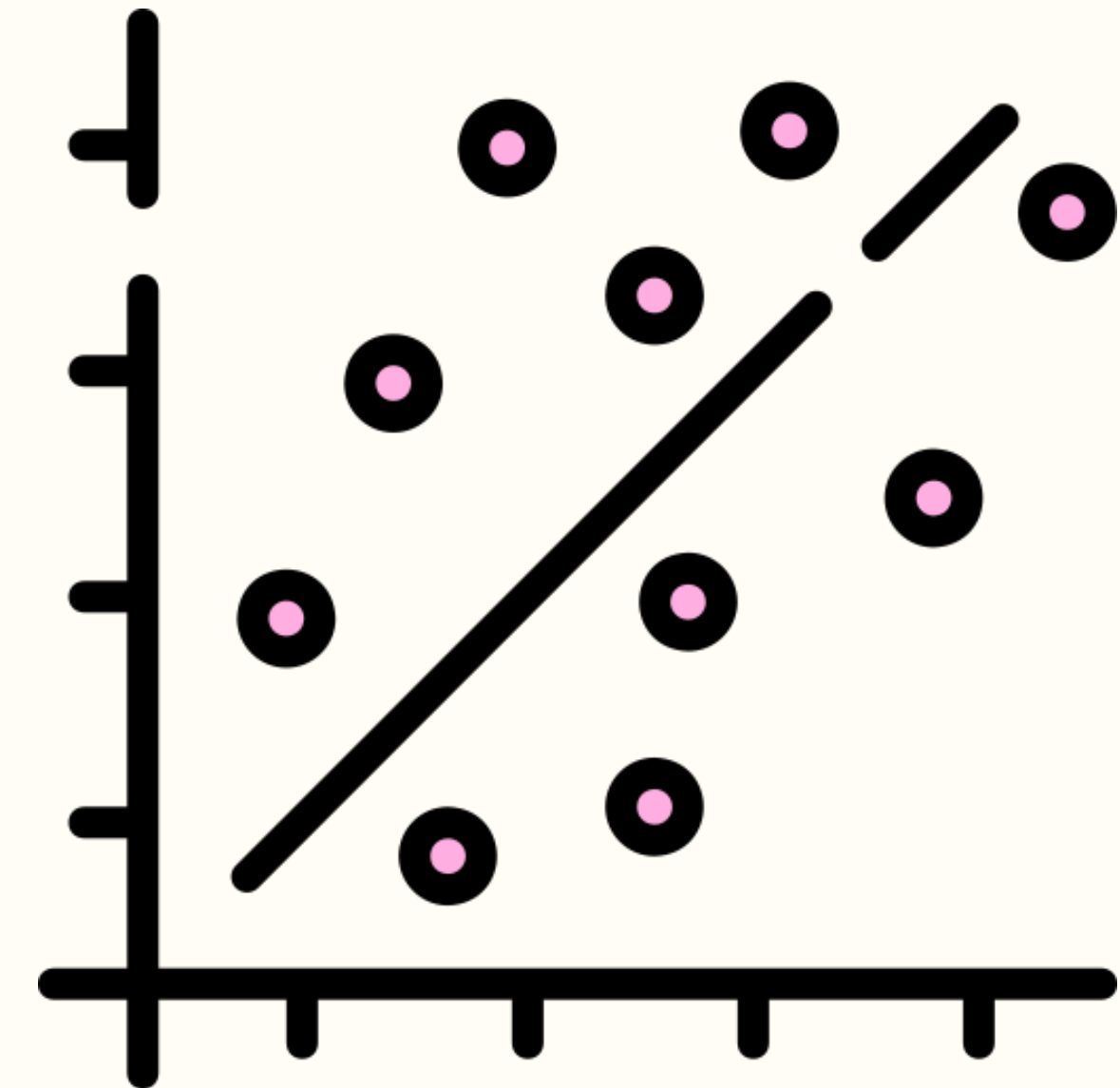
Simple linear regression

Heather Mattie, PhD

Linear regression

Like correlation analysis, **linear regression** can be used to investigate the **linear relationship** between two continuous random variables.

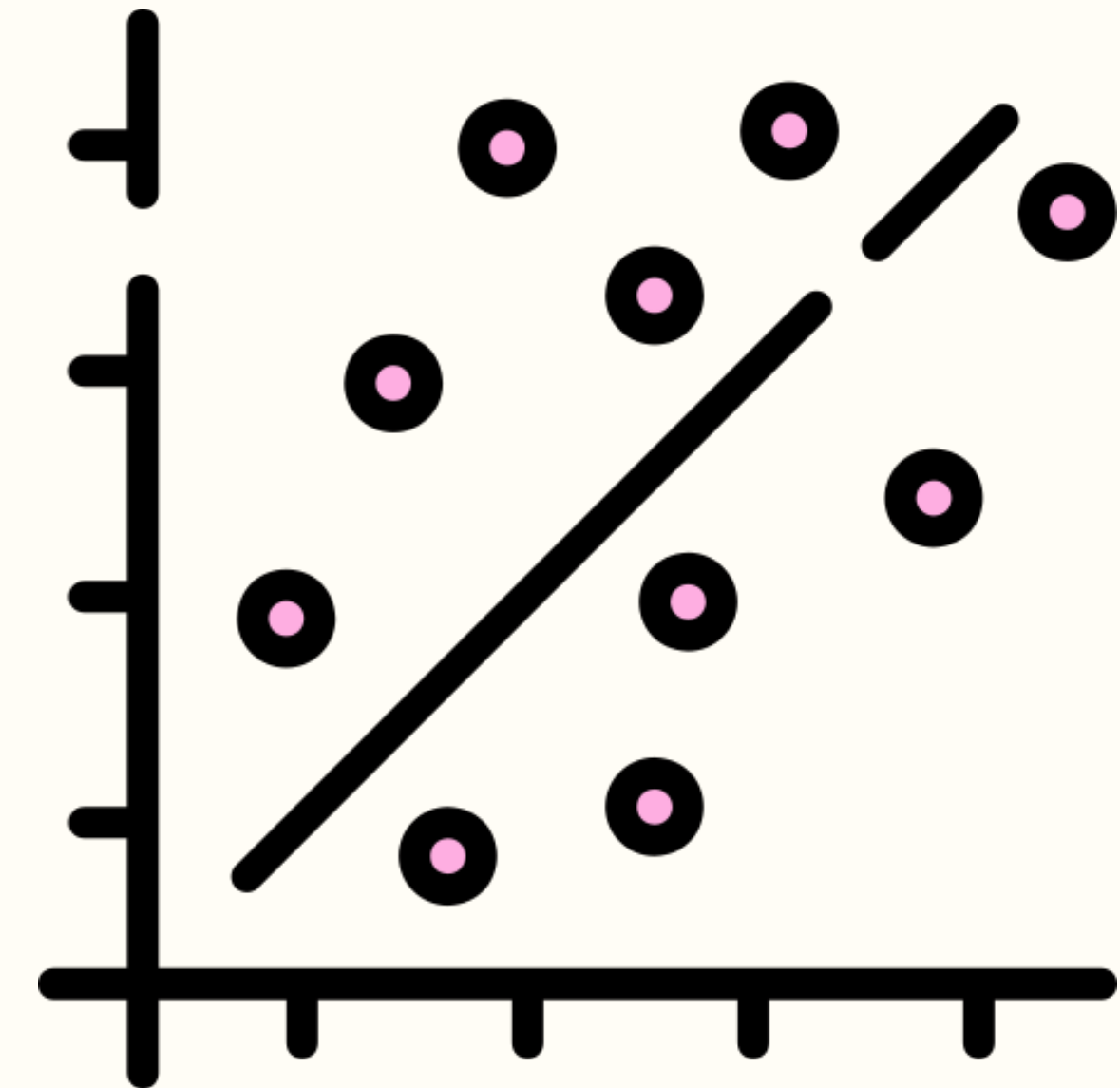
- Relationships between a continuous outcome variable and one or more risk factors that can be continuous, discrete, or categorical (nominal or ordinal), or some combination of these.



Linear regression

Linear regression can be used to

- **Estimate** or **predict** the value of a continuous outcome measurement given a set of risk factors
- Adjust for **confounders** when examining the relationship between an exposure and a continuous outcome
- Test for **effect measure modification**



Simple linear regression

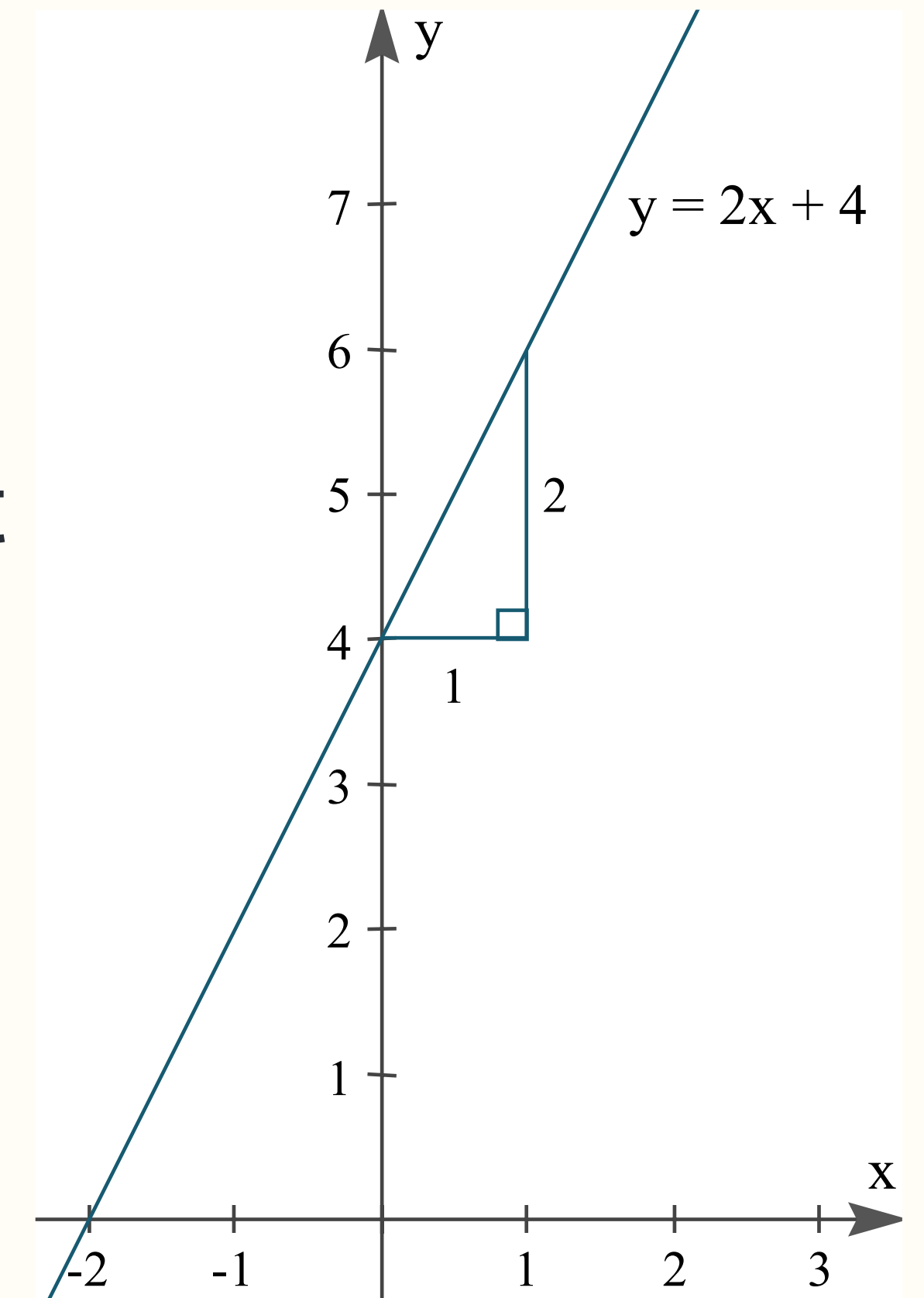
Simple linear regression can be used to explore the nature of the relationship between **two continuous variables**

- Regression looks at the change in one variable (**outcome** or **response** or **dependent variable**) that corresponds to a given change in the other variable (**explanatory variable** or **exposure** or **predictor** or **covariate** or **independent variable**)

$$y = mx + b$$

m: slope

b: y intercept



Assumptions

1. Linearity:

- The relationship between $\mu_{y|x}$ (mean of y given x) and x is linear

2. Normality:

- For a given value of x , y is normally distributed

3. Homoscedasticity:

- The standard deviation $\sigma_{y|x}$ is constant for all x

4. Independence:

- We have n independent pairs of observations

Example: Prenatal care

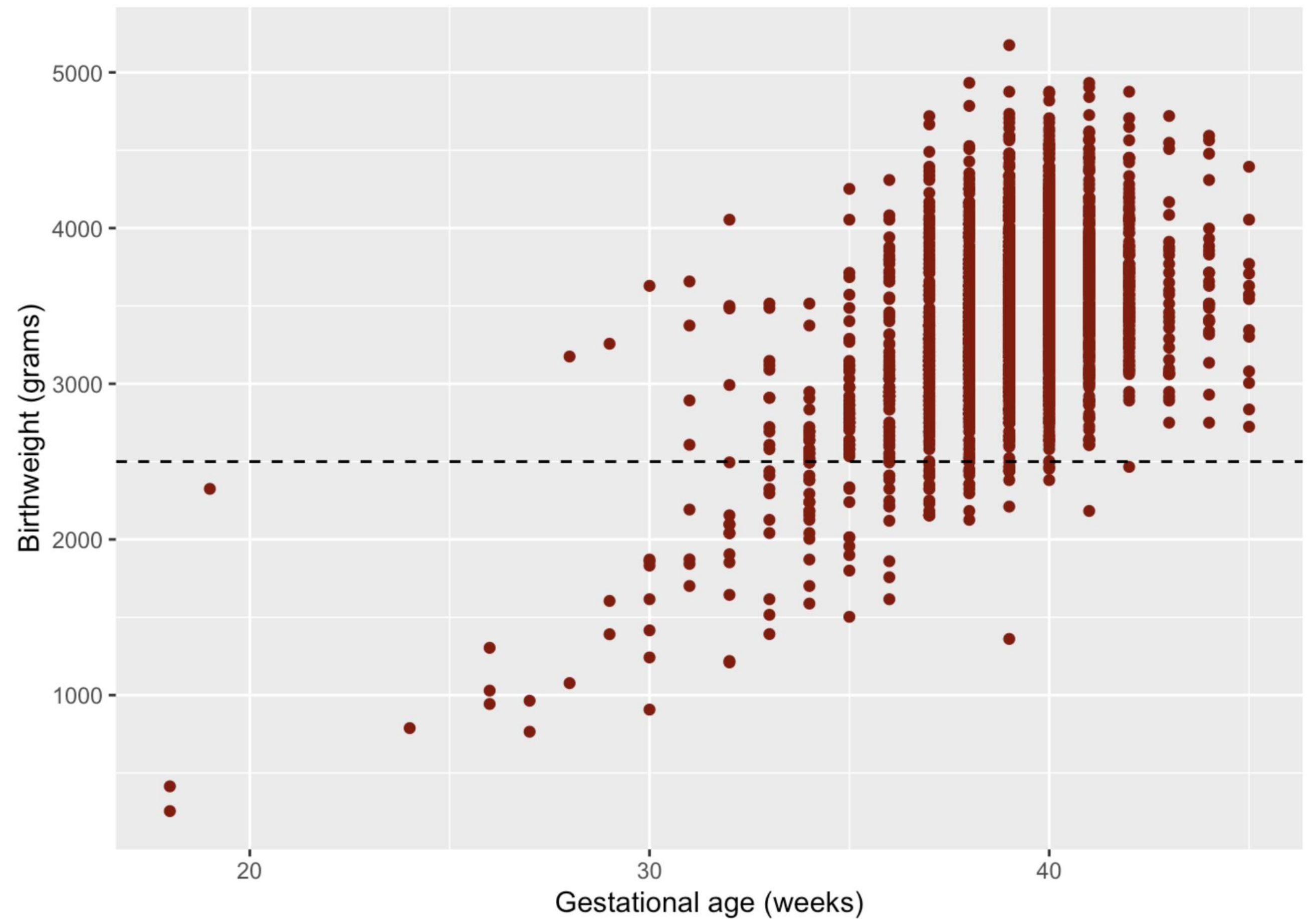
```
data <- read.table("KingCounty2001_data.txt", header = TRUE)
head(data)
```

```
##   gender plural age   race parity married  bwt smokeN drinkN firststep welfare
## 1      F      1  31 hispanic      4      1 3118      0      0      1      0
## 2      M      1  23   white      0      1 3466      0      0      1      0
## 3      F      1  24   black      1      0 3147      0      0      1      0
## 4      M      1  26 hispanic      2      1 3969      0      0      1      0
## 5      M      1  34   white      0      0 3005      0      0      1      0
## 6      M      1  15   black      0      0 2920      0      0      1      0
##   smoker drinker wpre wgain education gestation
## 1      N      N  122   22          5          40
## 2      N      N  160   50         12          39
## 3      N      N  150   38         13          40
## 4      N      N  175   15         12          39
## 5      N      N  123   17         17          40
## 6      N      N  180   12          9          36
```

Example: Prenatal care

Suppose the outcome is birthweight in grams and the exposure is gestational age in weeks.

- A linear regression model will quantify how birthweight and gestational age are associated



The **lm** function

The **lm** function stands for **linear model** and is used to fit a linear regression model

- The primary inputs are formula and data
- Simple linear regression formula:
outcome ~ predictor

```
m1 <- lm(bwt ~ gestation, data = data)
```

Regression line equation:

$$\mu_{y|x} = \beta_1 x + \beta_0$$

$$\mu_{\text{bwt}|\text{gestation}} = \beta_1(\text{gestation}) + \beta_0$$

The **lm** function

The **lm** function stands for **linear model** and is used to fit a linear regression model

- The summary function outputs details about the fit model

Fitted regression line equation:

Mean birthweight = $124.1(\text{gestation}) - 1410.7$

```
summary(m1)
```

```
##
## Call:
## lm(formula = bwt ~ gestation, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2068.46  -301.93    -9.56   273.44  1745.54
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1410.7      155.8   -9.055  <2e-16 ***
## gestation     124.1        4.0   31.026  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 475.3 on 2498 degrees of freedom
## Multiple R-squared:  0.2782, Adjusted R-squared:  0.2779
## F-statistic: 962.6 on 1 and 2498 DF,  p-value: < 2.2e-16
```

Coefficient interpretation

Fitted regression line equation:

Mean birthweight = $124.1(\text{gestation}) - 1410.7$

Slope interpretation: for every 1 week increase in gestational age, birthweight increases by 124.1 grams on average.

Intercept interpretation: the estimated mean birthweight when gestational age is 0 weeks is -1410.7 grams.

```
summary(m1)
```

```
##
## Call:
## lm(formula = bwt ~ gestation, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2068.46  -301.93    -9.56   273.44  1745.54
##
## Coefficients:
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```

Coefficient of determination

R^2 is the **coefficient of determination**, which quantifies the proportion of the variance in the dependent variable that is explained by the independent variable.

- Ranges from 0 to 1
- A higher R^2 value indicates a better fit model
- In simple linear regression, R^2 is the Pearson correlation coefficient squared

```
summary(m1)
```

```
##
## Call:
## lm(formula = bwt ~ gestation, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2068.46  -301.93    -9.56   273.44  1745.54
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```

Confidence interval for slope

The confidence interval for all of the coefficients of the model, including the intercept, can be calculated using the **confint** function

```
confint(m1)
```

##	2.5 %	97.5 %
## (Intercept)	-1716.1938	-1105.1835
## gestation	116.2624	131.9502

Hypothesis tests

The p-value of the slope coefficient is used to test if the relationship between the outcome and exposure is significant

- If the p-value is < 0.05 , the slope is not equal to 0 and there is a statistically significant relationship between the outcome and exposure

```
summary(m1)
```

```
##
## Call:
## lm(formula = bwt ~ gestation, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2068.46  -301.93    -9.56   273.44  1745.54
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```

Prediction

One important use of a linear regression model is prediction of a continuous outcome

- Example: the predicted birthweight of a baby with gestational age equal to 40 weeks is $124.1(40) - 1410.7 = 3,553.3$ grams.

```
summary(m1)
```

```
##
## Call:
## lm(formula = bwt ~ gestation, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2068.46  -301.93    -9.56   273.44  1745.54
##
## Coefficients:
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## gestation     124.1        4.0   31.026  <2e-16 ***
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```

Prediction

One important use of a linear regression model is prediction of a continuous outcome

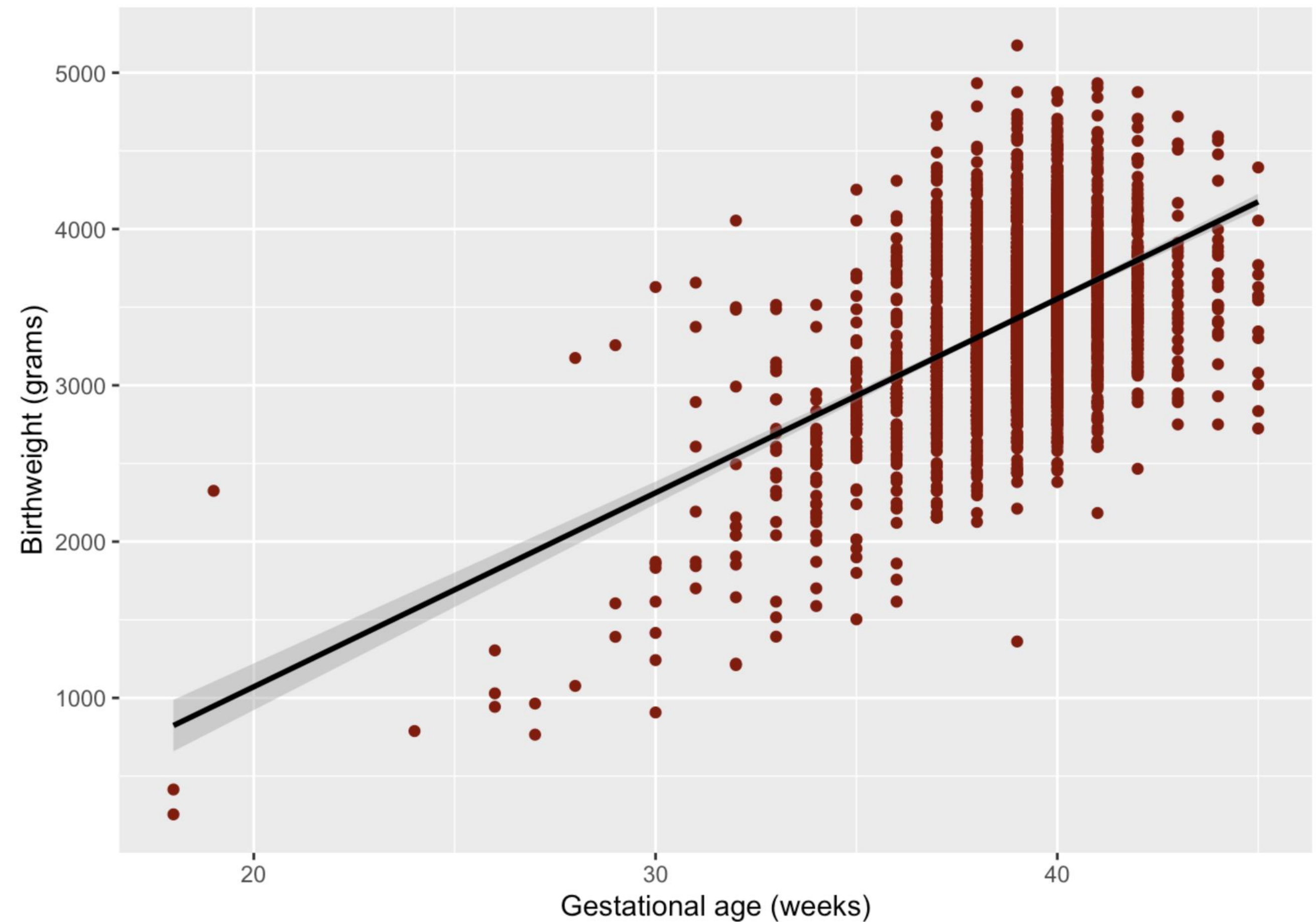
- Example: the predicted birthweight of a baby with gestational age equal to 28 weeks is $124.1(28) - 1410.7 = 2,064.1$ grams.

```
summary(m1)
```

```
##
## Call:
## lm(formula = bwt ~ gestation, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2068.46  -301.93    -9.56   273.44  1745.54
##
## Coefficients:
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## (Intercept)  -1410.7      155.8   -9.055  <2e-16 ***
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## Multiple R-squared:  0.2782, Adjusted R-squared:  0.2779
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```

Regression line plot

```
data %>%  
  ggplot(aes(gestation, bwt)) +  
  geom_point(color = "darkred") +  
  geom_smooth(method = "lm", se = TRUE, color = "black") +  
  xlab("Gestational age (weeks)") +  
  ylab("Birthweight (grams)")
```



Summary

- Simple linear regression quantifies the relationship between two continuous variables
- The **lm** function can be used to fit a linear regression model
- The **summary** function outputs details about the model, including coefficient values and R^2
- A linear regression model can be used for prediction
- A plot of the fitted regression line can be created using the **geom_smooth** geometry

Linear Regression I

Evaluation of model assumptions

Heather Mattie, PhD

Linear regression assumptions

1. Linearity:

- The relationship between $\mu_{y|x}$ (mean of y given x) and x is a straight line

2. Normality:

- For a given value of x , y is normally distributed

3. Homoscedasticity:

- The standard deviation $\sigma_{y|x}$ is constant for all x

4. Independence:

- We have n independent pairs of observations

Example: Prenatal care

```
data <- read.table("KingCounty2001_data.txt", header = TRUE)
head(data)
```

```
##   gender plural age   race parity married  bwt smokeN drinkN firststep welfare
## 1      F      1  31 hispanic      4      1 3118      0      0      1      0
## 2      M      1  23   white      0      1 3466      0      0      1      0
## 3      F      1  24   black      1      0 3147      0      0      1      0
## 4      M      1  26 hispanic      2      1 3969      0      0      1      0
## 5      M      1  34   white      0      0 3005      0      0      1      0
## 6      M      1  15   black      0      0 2920      0      0      1      0
##   smoker drinker wpre wgain education gestation
## 1      N      N  122   22          5          40
## 2      N      N  160   50         12          39
## 3      N      N  150   38         13          40
## 4      N      N  175   15         12          39
## 5      N      N  123   17         17          40
## 6      N      N  180   12          9          36
```

ggfortify and autoplot

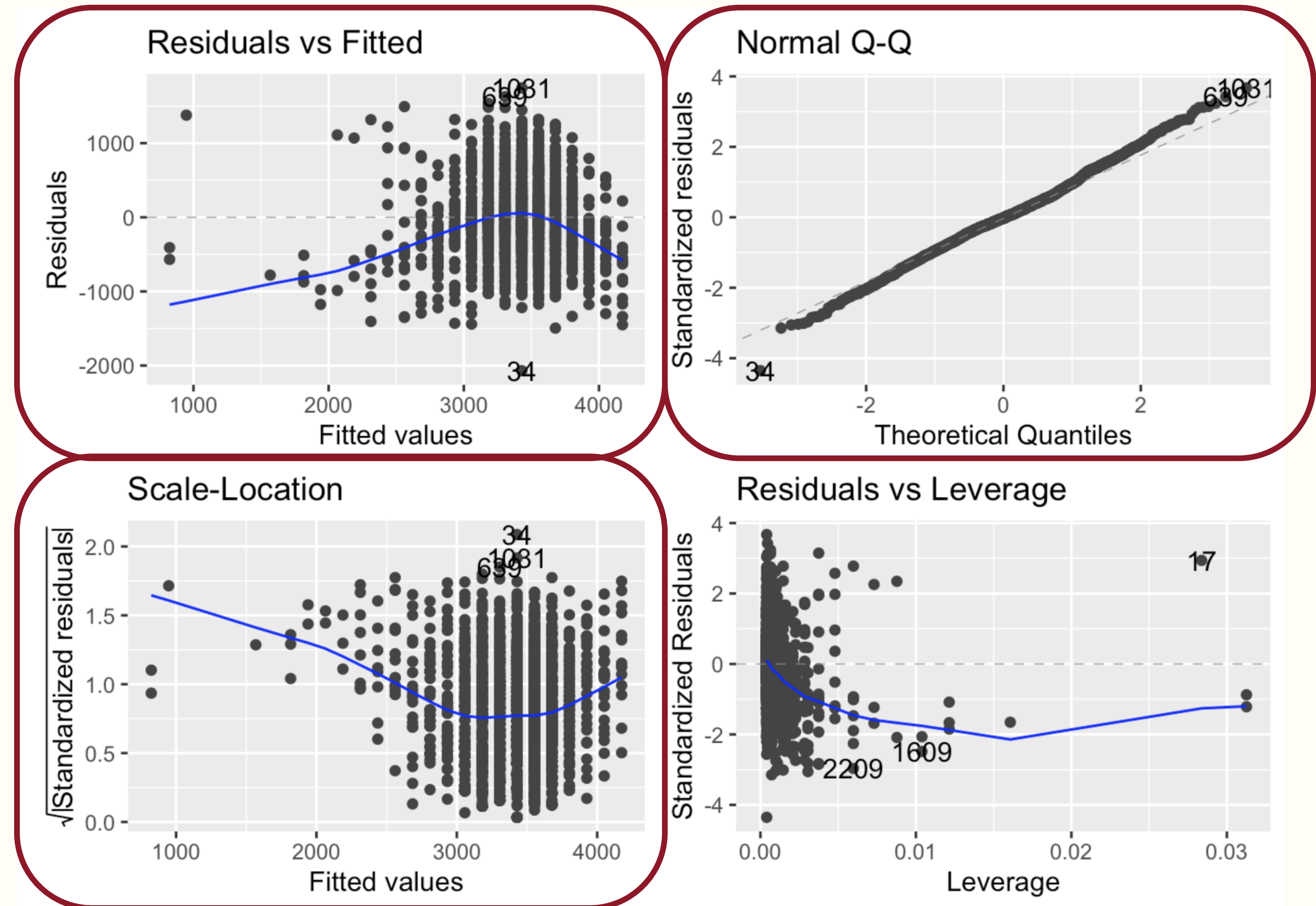
The function **autoplot** from the **ggfortify** package can be used to create plots to check if the assumptions of linear regression are met

```
library(ggfortify)  
autoplot(m1)
```

ggfortify and autoplot

The function **autoplot** from the **ggfortify** package can be used to create plots to check if the assumptions of linear regression are met

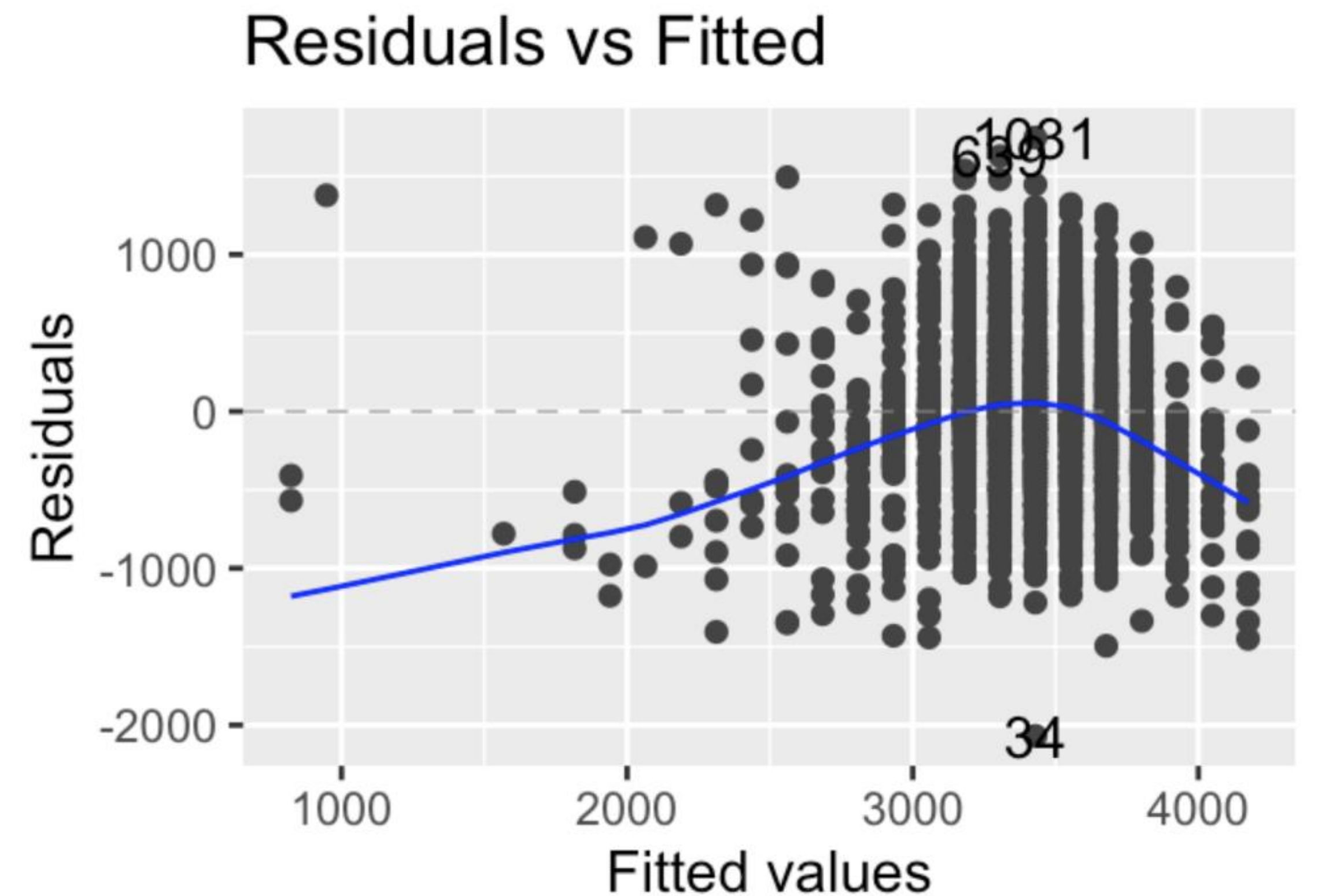
- Four plots are created, we will use 3 of those plots



Linearity

A plot of the residuals (difference between the actual values and the values predicted with the regression line) versus the fitted values is used to check **linearity**.

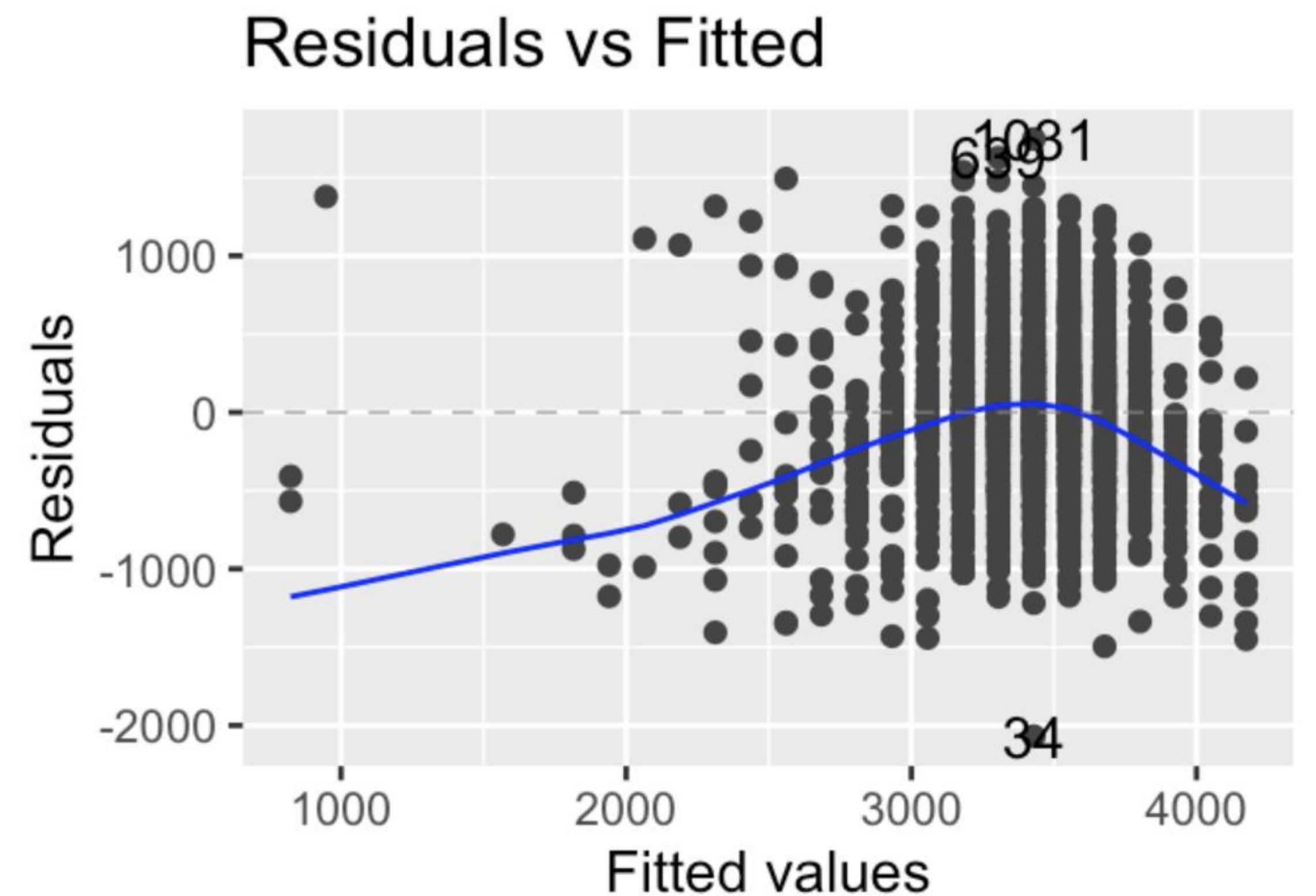
- A horizontal line, without distinct patterns is an indication for a linear relationship



Linearity

The linearity assumption is violated in our dataset

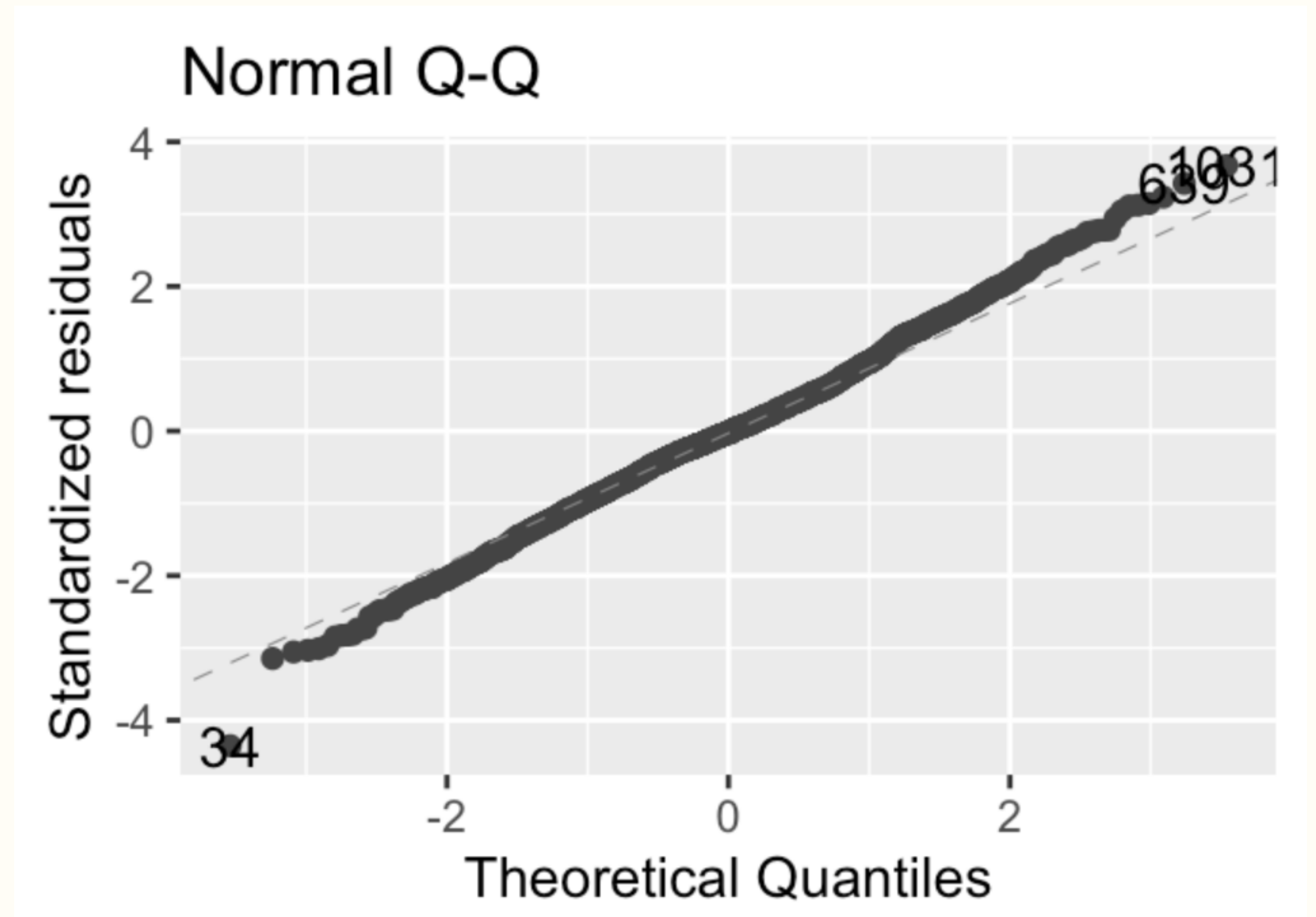
- Due to the relatively small amount of data with lower values of gestational age
- A potential solution would be to transform the exposure variable (log, square root, etc.)



Normality

A quantile-quantile plot (Normal Q-Q plot) is used to check the **normality** assumption.

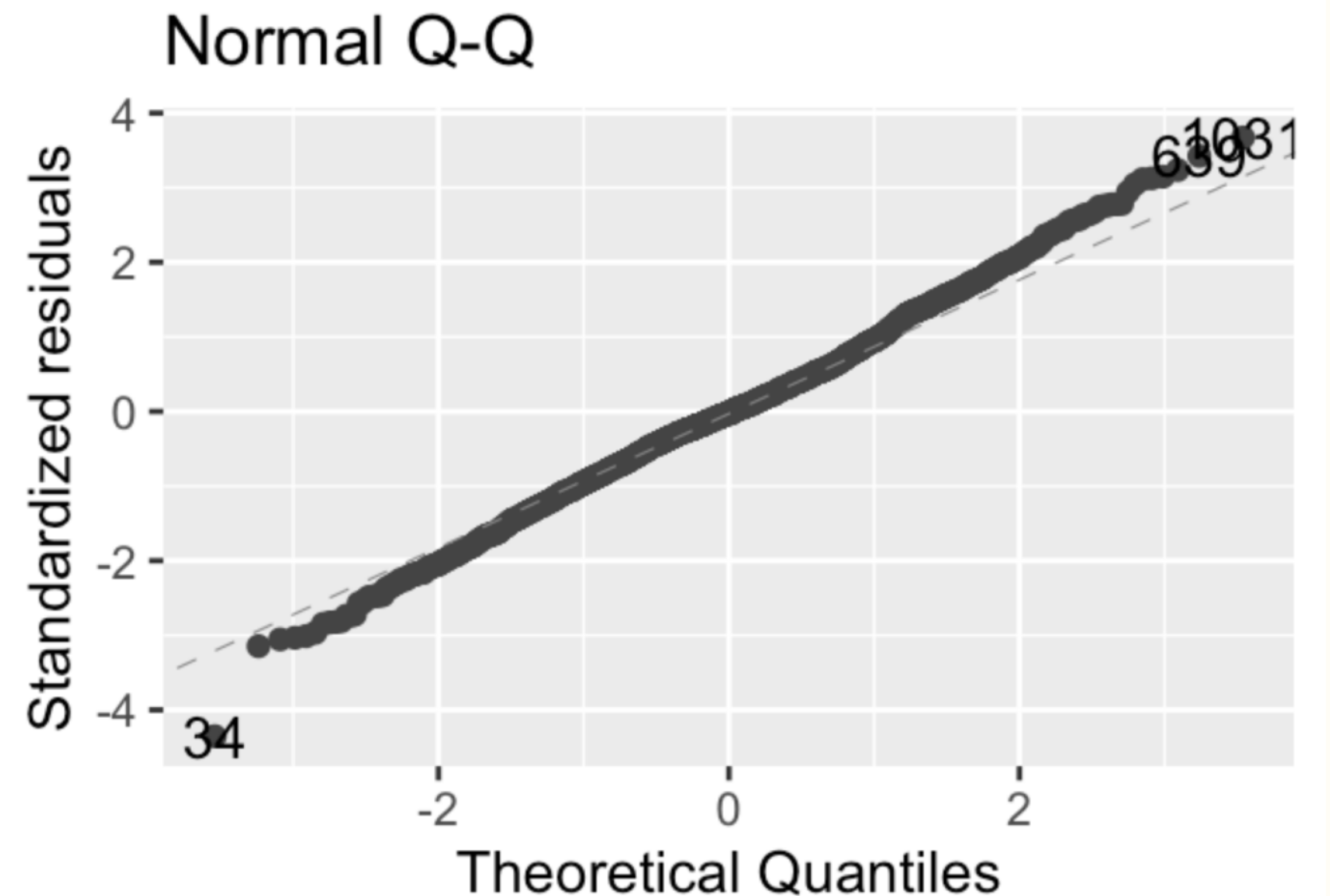
- The normality assumption is met if the standardized residuals follow the straight dashed line



Normality

Looking at the plot, it's difficult to be sure the normality assumption is met.

- Can use the Shapiro-Wilk test to test the normality assumption



Normality

The normality assumption is not met in the dataset since the p-value is < 0.05 .

A few potential solutions include

- Applying data transformations to the exposure variable
- Using robust regression methods that are less sensitive to non-normal data or outliers
- Remove outliers

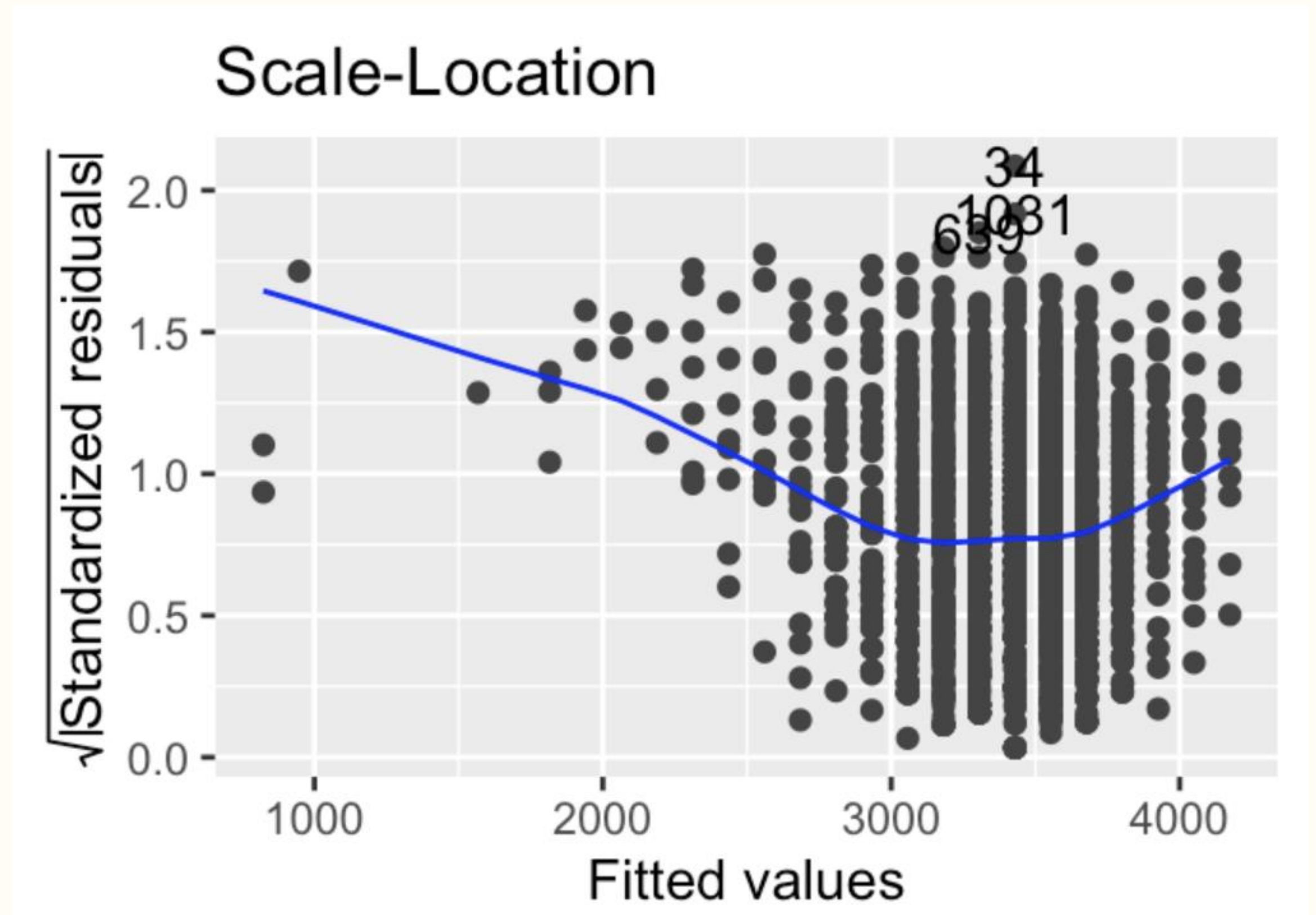
```
shapiro.test(residuals(m1))
```

```
##  
## Shapiro-Wilk normality test  
##  
## data: residuals(m1)  
## W = 0.99666, p-value = 2.551e-05
```

Homoscedasticity

A plot of the square root of the standardized residuals versus the fitted values is used to check the **homoscedasticity** assumption.

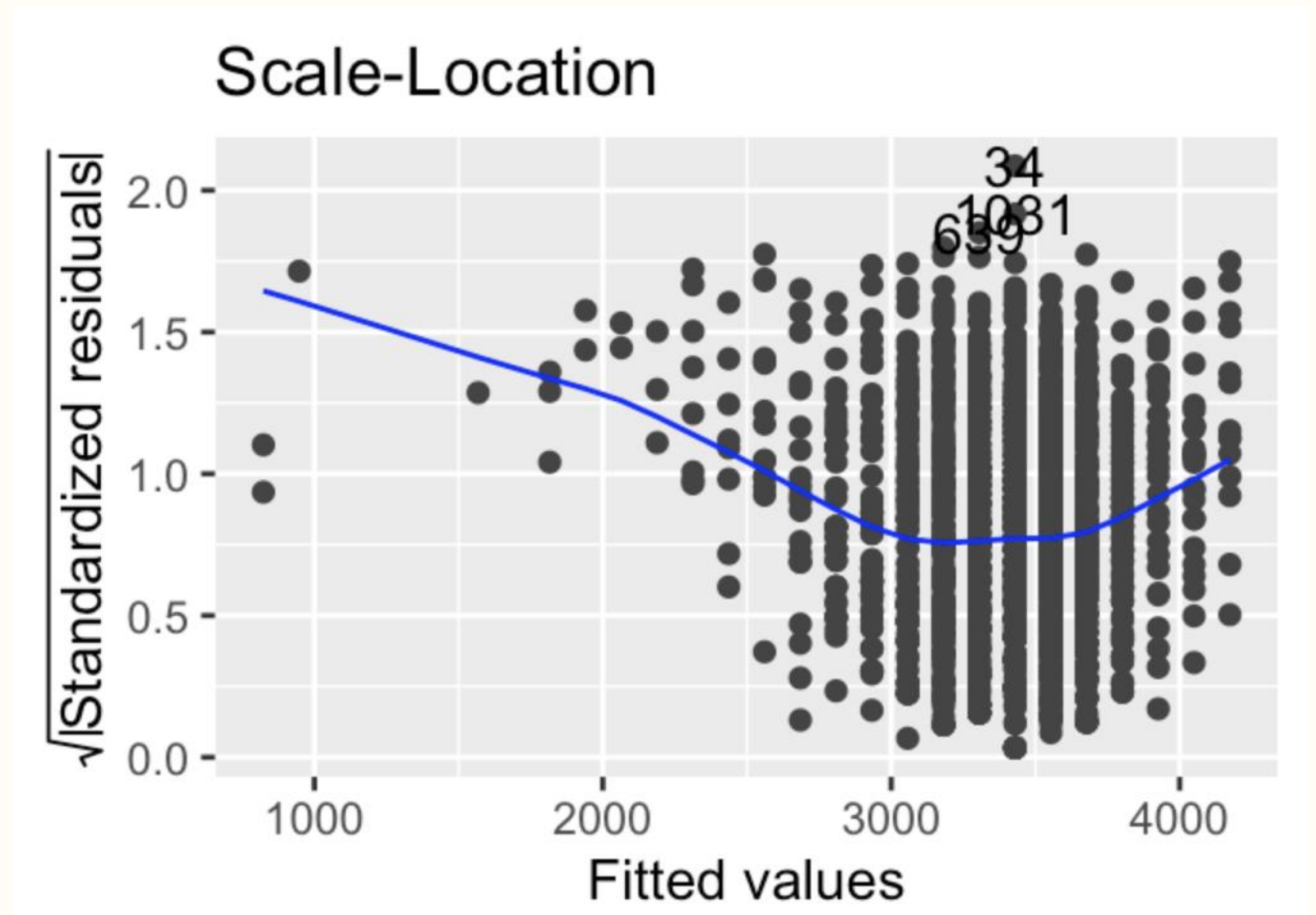
- Called a scale-location or spread-location plot
- A horizontal line with equally spread points is a good indication of homoscedasticity.



Homoscedasticity

The homoscedasticity assumption is violated in our dataset.

- Due to the relatively small amount of data with lower values of gestational age
- A potential solution would be to transform the outcome variable (log, square root, etc.)



Independence

The only way to check the **independence** assumption is having knowledge of the study / sampling design.

- Contact those responsible for data collection
- Read study documentation
- Create a subset of the original dataset that only includes independent pairs of observations

The independence assumption holds for this dataset.

Summary

- The assumptions of linear regression should be checked before choosing and interpreting a final model
- The linearity assumption can be checked with a plot of the residuals versus the fitted values
- The normality assumption can be checked with a Normal Q-Q plot or Shapiro-Wilk test
- The homoscedasticity assumption can be checked with a plot of the square root of the standardized residuals versus the fitted values
- The independence assumption can only be checked by verifying the study / sampling design for data collection

Linear Regression II

Heather Mattie, PhD



Linear Regression II Roadmap

1. Multivariate linear regression
2. Collinearity
3. Model selection

Linear Regression II

Multivariate linear regression

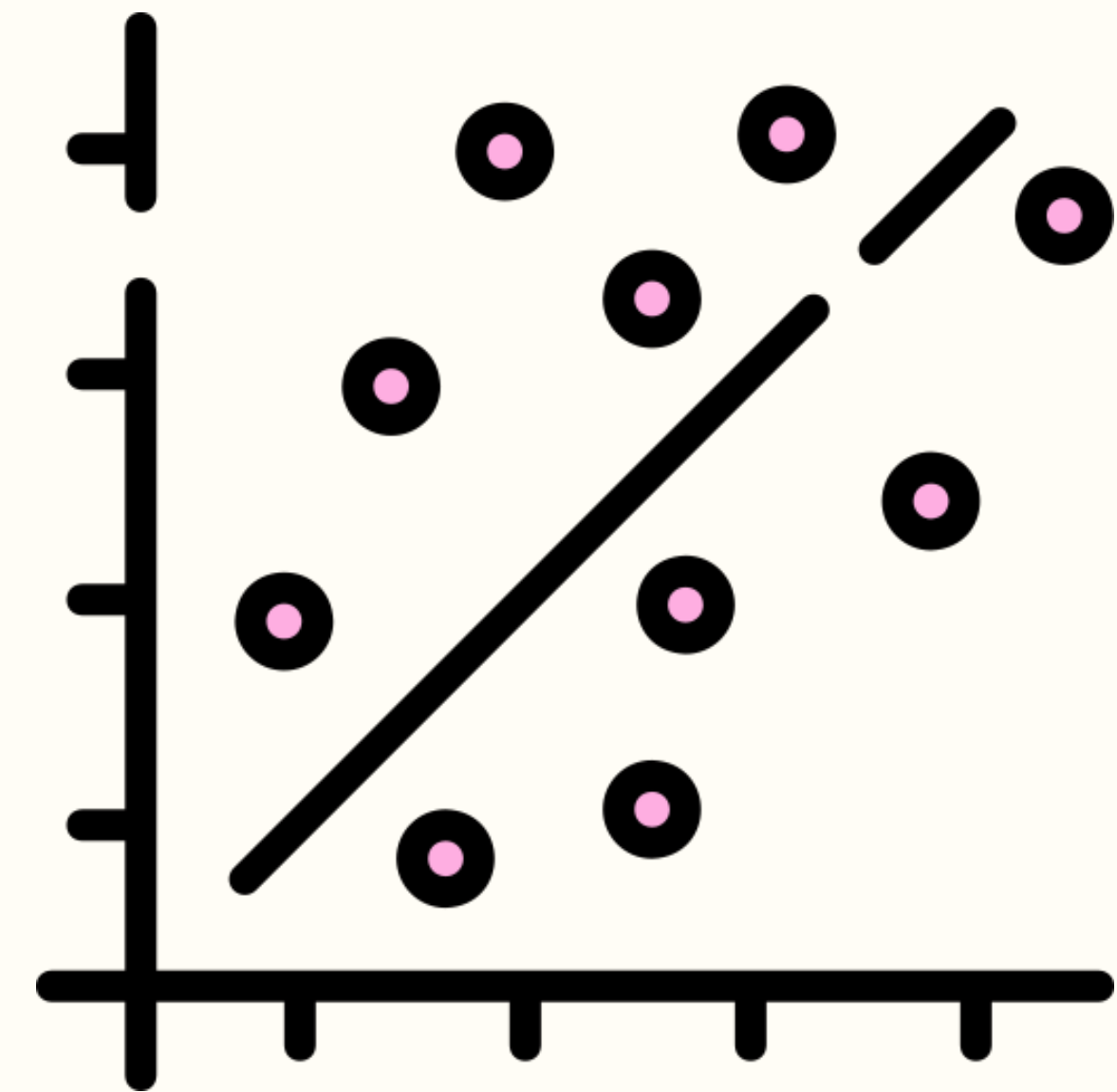
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Multivariate linear regression

Multivariate linear regression can be used to explore more complex relationships involving a **continuous outcome** and **two or more explanatory variables**.

Regression line equation:

$$\mu_{y|x} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_q x_q$$



Assumptions

1. Linearity:

- The relationship between $\mu_{y|\mathbf{x}}$ (mean of y given $x_1, x_2, \dots, x_q$) and $\mathbf{x}$ is linear

2. Normality:

- For a fixed set of values for $x_1, x_2, \dots, x_q$, y is normally distributed

3. Homoscedasticity:

- The standard deviation $\sigma_{y|\mathbf{x}}$ is constant across all values of the explanatory variables

4. Independence:

- The observations are independent

Example: Prenatal care

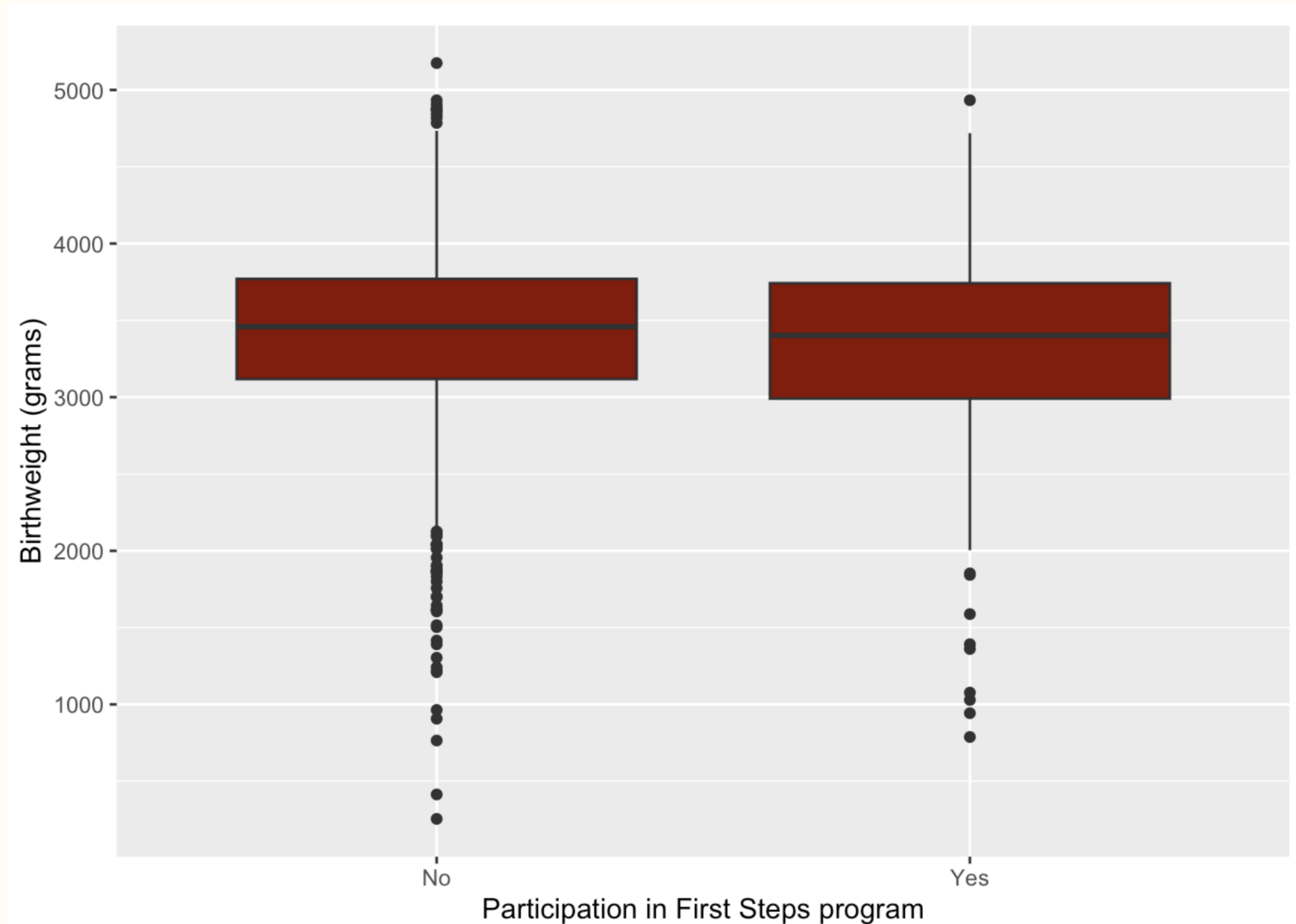
```
data <- read.table("KingCounty2001_data.txt", header = TRUE)
head(data)
```

```
##   gender plural age   race parity married  bwt smokeN drinkN firststep welfare
## 1      F      1  31 hispanic      4      1 3118      0      0      1      0
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## 6      M      1  15   black      0      0 2920      0      0      1      0
##   smoker drinker wpre wgain education gestation
## 1      N      N  122   22          5          40
## 2      N      N  160   50         12          39
## 3      N      N  150   38         13          40
## 4      N      N  175   15         12          39
## 5      N      N  123   17         17          40
## 6      N      N  180   12          9          36
```

Example: Prenatal care

Suppose the outcome is birthweight in grams and the exposure is participation in the First Steps program.

```
data1 %>% ggplot(aes(as.factor(firststep_cat), bwt)) +  
  geom_boxplot(fill = "darkred") +  
  ylab("Birthweight (grams)") +  
  xlab("Participation in First Steps program")
```



Crude model

A linear regression model with one explanatory variable (exposure) of interest is called a **crude model**.

- No other covariates have been “adjusted” for, meaning no other covariates have been added to the model yet

```
m <- lm(bwt ~ firststep, data = data)
summary(m)
```

```
##
## Call:
## lm(formula = bwt ~ firststep, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3169.7  -306.9    34.3   345.3  1750.3
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3424.69     12.21  280.581  <2e-16 ***
## firststep     -66.18     30.40   -2.177   0.0296 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 558.9 on 2498 degrees of freedom
## Multiple R-squared:  0.001894,    Adjusted R-squared:  0.001494
## F-statistic: 4.739 on 1 and 2498 DF,  p-value: 0.02958
```

Crude model

Mean **bwt** = $-66.18(\text{firststep} = 1) + 3424.69$

- **firststep** is called an indicator variable

Interpretation:

- Infants of participants in the First Steps program weighed, on average, 66.18 grams less at birth.

Mean bwt for program participants: 3358.51

Mean bwt for non-participants: 3424.69

```
m <- lm(bwt ~ firststep, data = data)
summary(m)
```

```
##
## Call:
## lm(formula = bwt ~ firststep, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3169.7  -306.9    34.3   345.3  1750.3
##
## Coefficients:
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## (Intercept)   3424.69     12.21  280.581  <2e-16 ***
## firststep     -66.18     30.40   -2.177   0.0296 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 558.9 on 2498 degrees of freedom
## Multiple R-squared:  0.001894,    Adjusted R-squared:  0.001494
## F-statistic: 4.739 on 1 and 2498 DF,  p-value: 0.02958
```

Adjusted model

When other covariates are added to the crude model, the model becomes an “adjusted model”.

- Interpretations of coefficients change slightly

```
m1 <- lm(bwt ~ firststep + gestation, data = data)
summary(m1)
```

```
##
## Call:
## lm(formula = bwt ~ firststep + gestation, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2033.17  -300.12   -13.39   276.11  1738.79
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1396.289    155.998  -8.951  <2e-16 ***
## firststep    -42.037     25.856  -1.626    0.104
## gestation    123.910      4.001   30.973  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 475.2 on 2497 degrees of freedom
## Multiple R-squared:  0.2789, Adjusted R-squared:  0.2783
## F-statistic: 482.9 on 2 and 2497 DF, p-value: < 2.2e-16
```

Adjusted model

Interpretations:

- Infants of participants in the First Steps program weighed, on average, 42.037 grams less at birth, adjusting for gestational age.
- For every 1 week increase in gestation, birthweight increases by 123.91 grams on average, holding all other covariates constant.

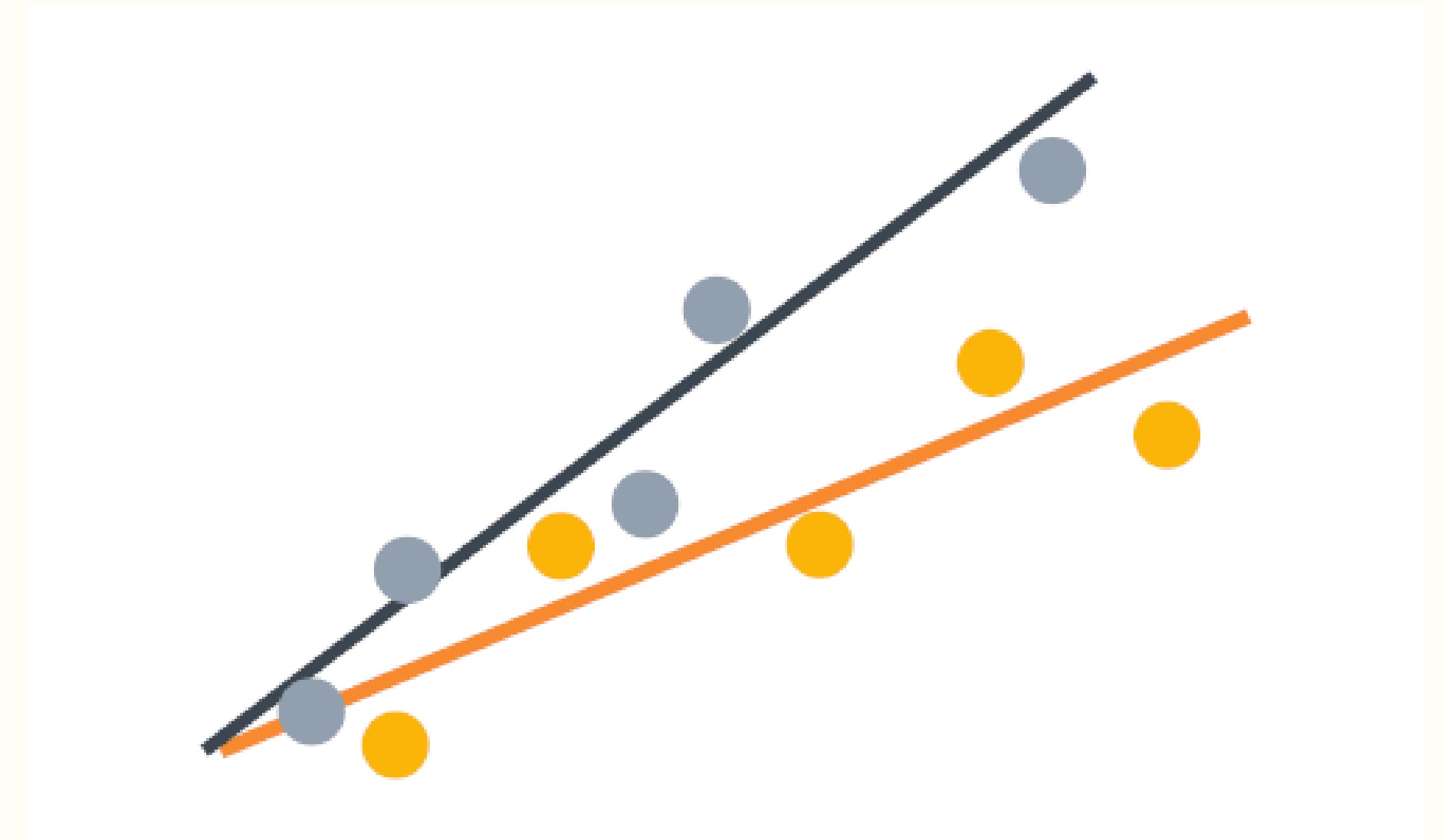
```
m1 <- lm(bwt ~ firststep + gestation, data = data)
summary(m1)
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```
##
## Call:
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## firststep    -42.037     25.856  -1.626    0.104
## gestation    123.910      4.001   30.973  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 475.2 on 2497 degrees of freedom
## Multiple R-squared:  0.2789, Adjusted R-squared:  0.2783
## F-statistic: 482.9 on 2 and 2497 DF, p-value: < 2.2e-16
```

Interaction term

Suppose we wanted to test if the relationship between birthweight and gestational age is different for infants whose mothers participated in the First Steps program and those whose mothers did not participate

- This is testing effect measure modification
- Create an **interaction term** to test this



Interaction term

In R, the syntax

covariate₁ * covariate₂

fits a model with terms covariate₁,
covariate₂, and covariate₁ *
covariate₂ in it.

```
m2 <- lm(bwt ~ firststep*gestation, data = data)
summary(m2)
```

```
##
## Call:
## lm(formula = bwt ~ firststep * gestation, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2032.04  -300.60   -11.24   277.69  1738.69
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1440.077    176.788  -8.146 5.89e-16 ***
## firststep      154.516    374.075   0.413  0.680
## gestation     125.036     4.536  27.565 < 2e-16 ***
## firststep:gestation -5.072     9.629  -0.527  0.598
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 475.2 on 2496 degrees of freedom
## Multiple R-squared:  0.279, Adjusted R-squared:  0.2781
## F-statistic: 322 on 3 and 2496 DF, p-value: < 2.2e-16
```

Interaction term

The p-value for the interaction term is not <0.05 , so it should not be kept in the model.

- No evidence of effect measure modification

```
m2 <- lm(bwt ~ firststep*gestation, data = data)
summary(m2)
```

```
##
## Call:
## lm(formula = bwt ~ firststep * gestation, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2032.04  -300.60   -11.24    277.69   1738.69
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1440.077    176.788  -8.146 5.89e-16 ***
## firststep      154.516    374.075   0.413  0.680
## gestation     125.036     4.536  27.565 < 2e-16 ***
## firststep:gestation -5.072     9.629  -0.527  0.598
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 475.2 on 2496 degrees of freedom
## Multiple R-squared:  0.279, Adjusted R-squared:  0.2781
## F-statistic: 322 on 3 and 2496 DF, p-value: < 2.2e-16
```

Summary

- A linear regression model with two or more explanatory variables is multivariate linear regression.
- Multivariate linear regression can be used to evaluate more complex relationships involving an outcome and explanatory variables.
- An interaction term is used to test if a covariate is an effect measure modifier.

Linear Regression II

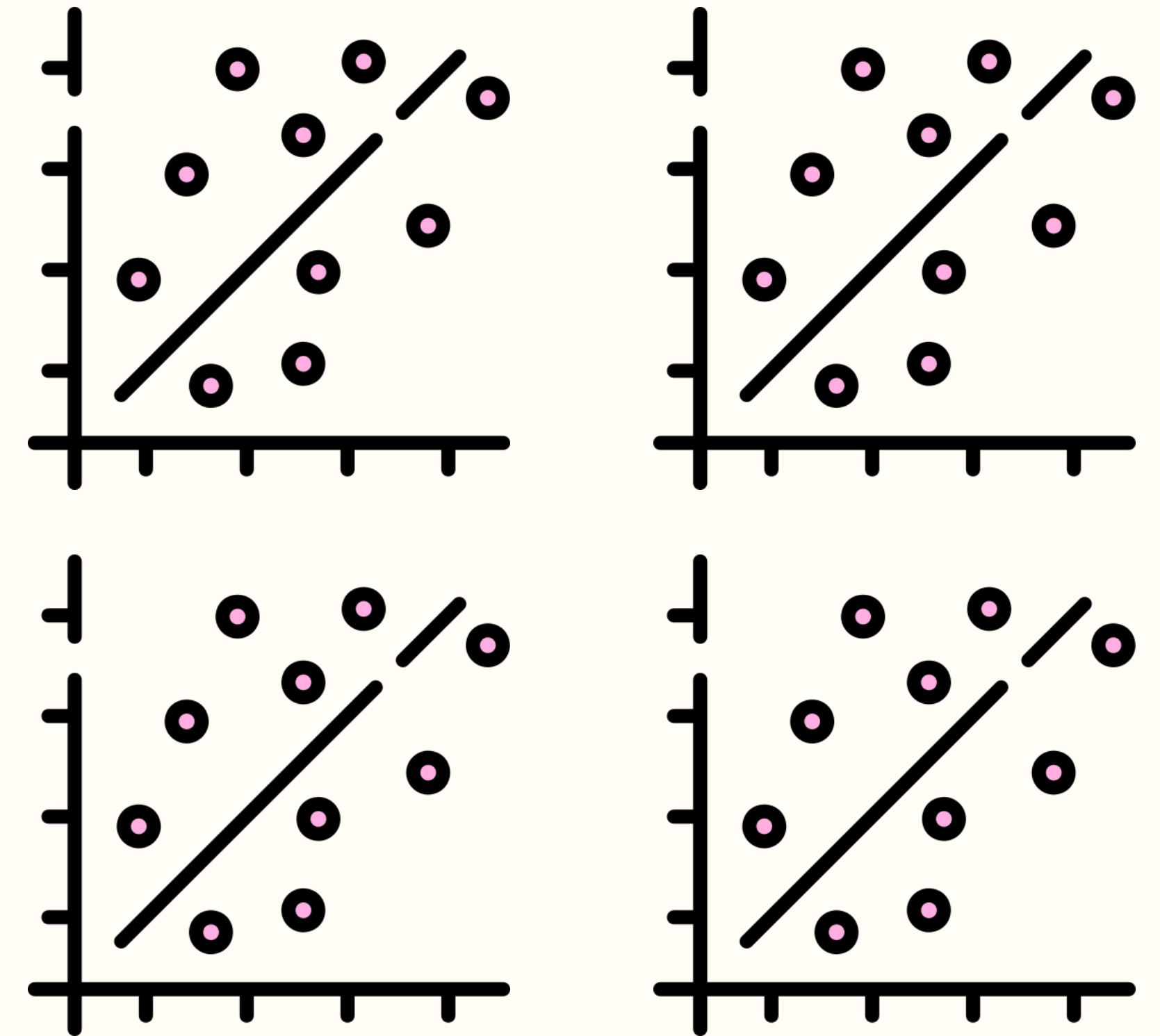
Collinearity

Heather Mattie, PhD

Collinearity

Collinearity occurs when independent variables are highly correlated.

- Difficult for the model to estimate individual effects on the outcome variable
- Inflates variance of the regression coefficients leading to unstable estimates and wider CIs
- Often, a decision about which to include in the final model is needed



Example: Prenatal care

```
data <- read.table("KingCounty2001_data.txt", header = TRUE)
head(data)
```

```
##   gender plural age   race parity married  bwt smokeN drinkN firststep welfare
## 1      F      1  31 hispanic      4      1 3118      0      0      1      0
## 2      M      1  23   white      0      1 3466      0      0      1      0
## 3      F      1  24   black      1      0 3147      0      0      1      0
## 4      M      1  26 hispanic      2      1 3969      0      0      1      0
## 5      M      1  34   white      0      0 3005      0      0      1      0
## 6      M      1  15   black      0      0 2920      0      0      1      0
##   smoker drinker wpre wgain education gestation
## 1      N      N  122   22          5          40
## 2      N      N  160   50         12          39
## 3      N      N  150   38         13          40
## 4      N      N  175   15         12          39
## 5      N      N  123   17         17          40
## 6      N      N  180   12          9          36
```

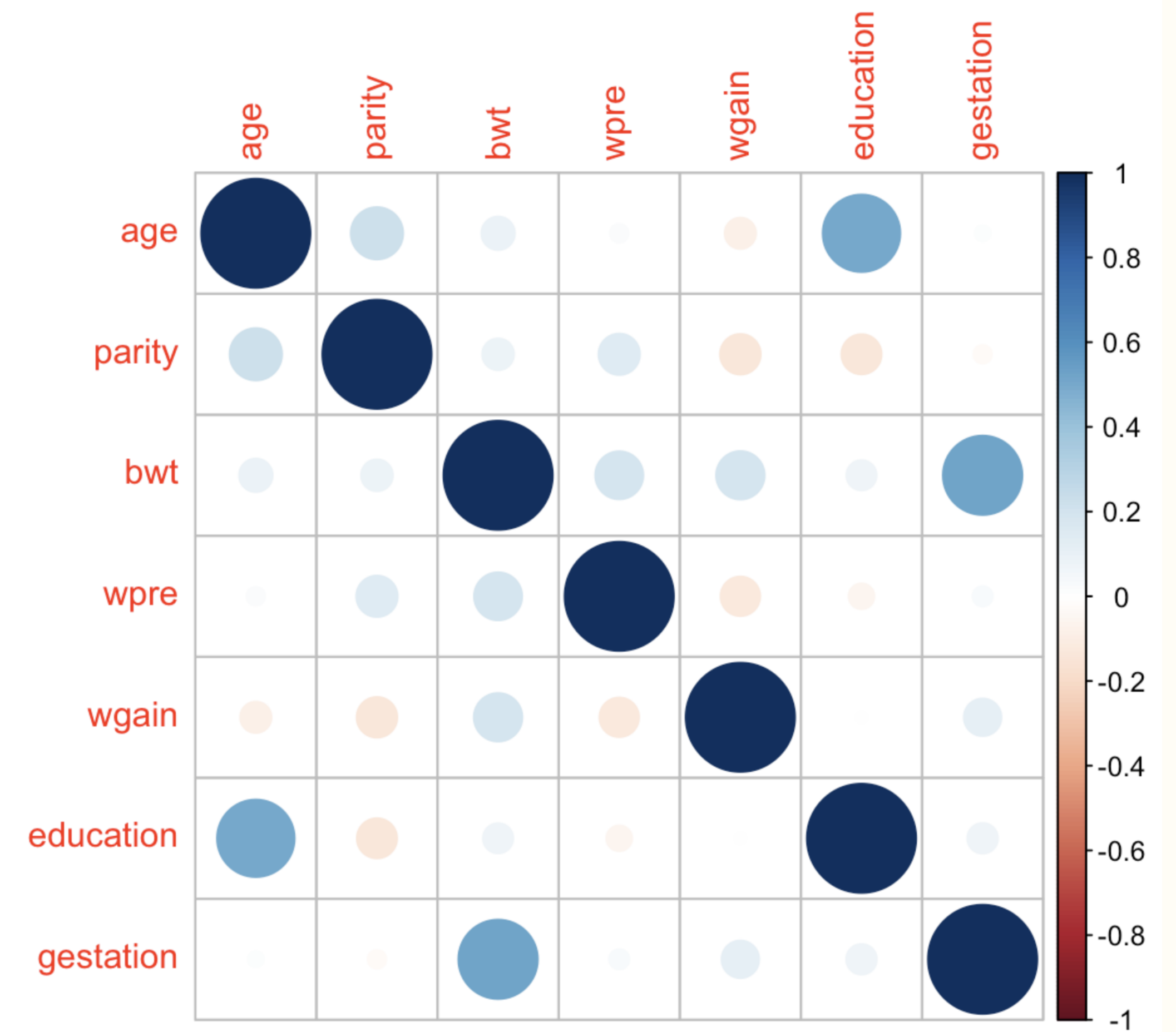
Correlation matrix

Collinearity is usually checked with a **correlation matrix**.

- Only for continuous covariates
- Can use Pearson or Spearman

```
vars <- c("age", "parity", "bwt", "wpre", "wgain", "education", "gestation")
data_subset <- data_cor[, vars]
corr_matrix <- cor(data_subset, method = "spearman")
```

```
corrplot(corr_matrix)
```

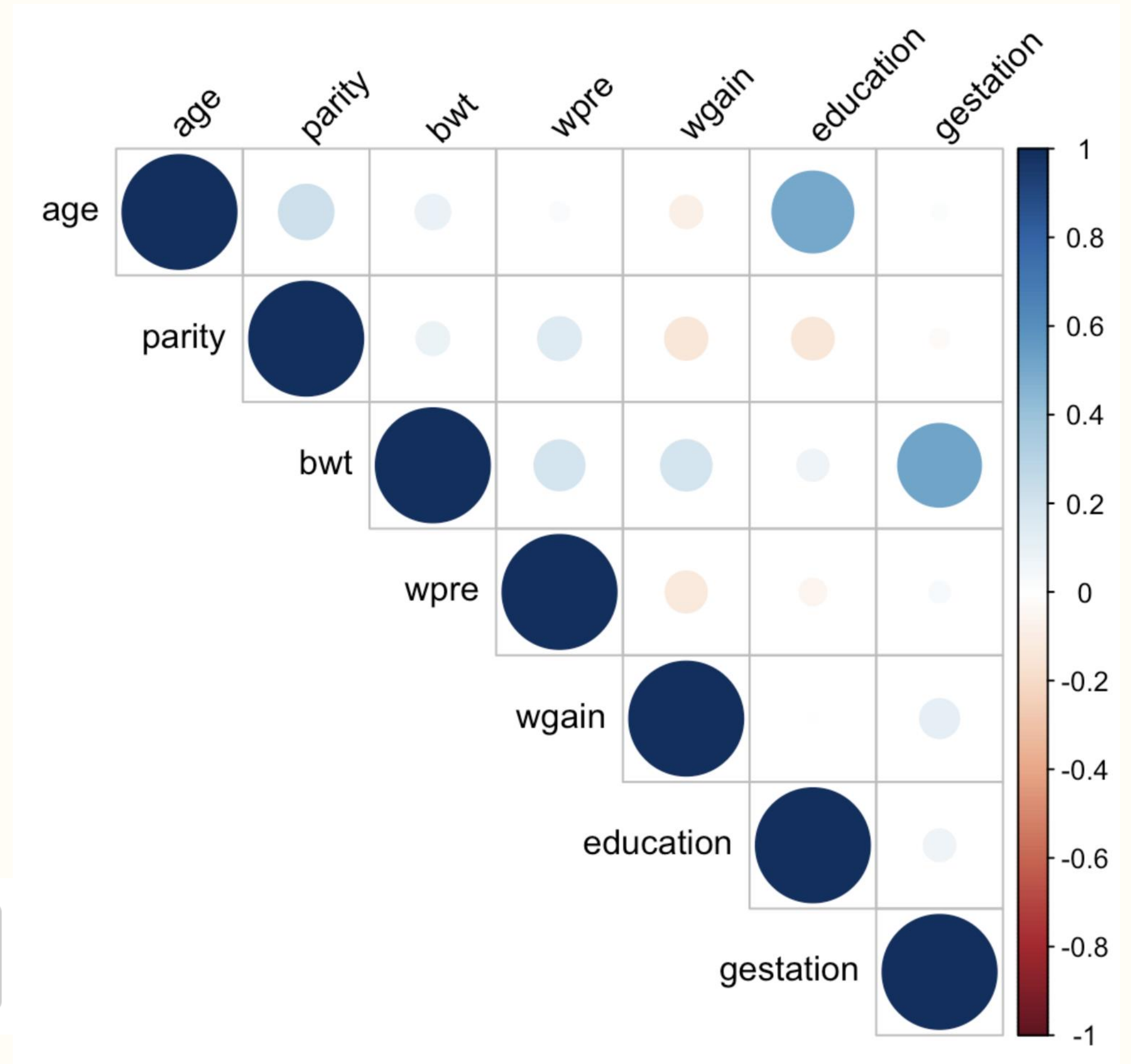


Correlation matrix

Collinearity is usually checked with a **correlation matrix**.

- Only for continuous covariates
- Can use Pearson or Spearman

```
corrplot(corr_matrix, type = "upper", tl.col = "black", tl.srt = 45)
```

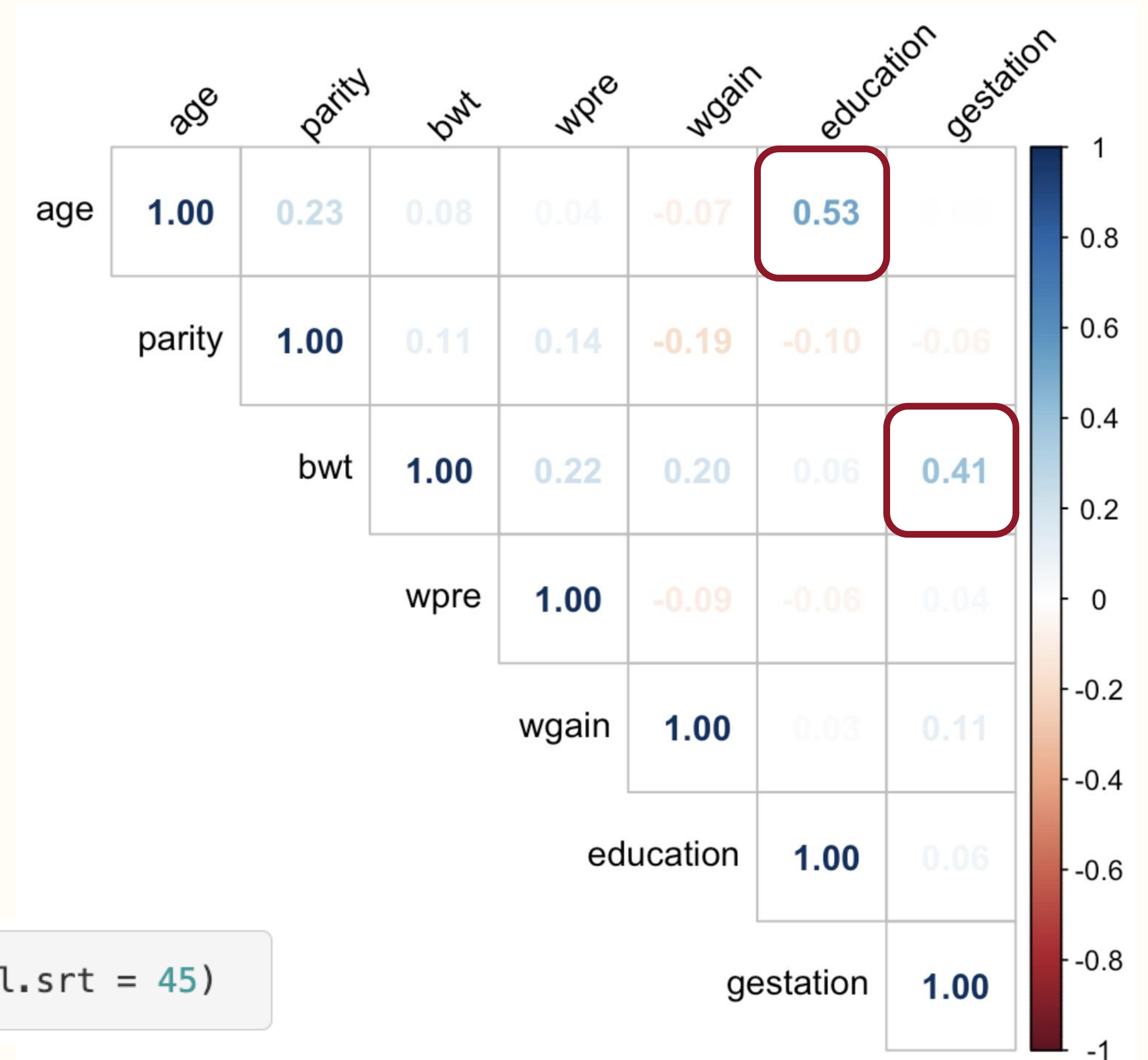


Correlation matrix

Collinearity is usually checked with a **correlation matrix**.

- Only for continuous covariates
- Can use Pearson or Spearman

```
corrplot(corr_matrix, method = "number", type = "upper", tl.col = "black", tl.srt = 45)
```



Correlation matrix

Collinearity is usually checked with a **correlation matrix**.

- Only for continuous covariates
- Can use Pearson or Spearman

```
library(Hmisc)
```

```
data_matrix <- as.matrix(data_subset)  
corr_res <- rcorr(data_matrix, type = "spearman")  
print(round(corr_res$r, digits = 2))
```

```
##          age parity  bwt  wpre wgain education gestation  
## age      1.00   0.23 0.08  0.04 -0.07      0.53      0.00  
## parity   0.23   1.00 0.11  0.14 -0.19     -0.10     -0.06  
## bwt      0.08   0.11 1.00  0.22  0.20      0.06      0.41  
## wpre     0.04   0.14 0.22  1.00 -0.09     -0.06      0.04  
## wgain    -0.07  -0.19 0.20 -0.09  1.00      0.03      0.11  
## education 0.53  -0.10 0.06 -0.06  0.03      1.00      0.06  
## gestation 0.00  -0.06 0.41  0.04  0.11      0.06      1.00
```

Correlation tests

The correlation value often isn't enough and a test for each pair of covariates should be performed to see which correlations are significant.

```
print(round(corr_res$P, digits = 3))
```

##		age	parity	bwt	wpre	wgain	education	gestation
##	age	NA	0.000	0.000	0.065	0.000	0.000	0.835
##	parity	0.000	NA	0.000	0.000	0.000	0.000	0.006
##	bwt	0.000	0.000	NA	0.000	0.000	0.002	0.000
##	wpre	0.065	0.000	0.000	NA	0.000	0.004	0.075
##	wgain	0.000	0.000	0.000	0.000	NA	0.187	0.000
##	education	0.000	0.000	0.002	0.004	0.187	NA	0.002
##	gestation	0.835	0.006	0.000	0.075	0.000	0.002	NA

Correlation tests

The correlation value often isn't enough and a test for each pair of covariates should be performed to see which correlations are significant.

- The boxed values show non-significant p-values (variables are not collinear)

```
print(round(corr_res$P, digits = 3))
```

##		age	parity	bwt	wpre	wgain	education	gestation
##	age							
##	parity	0.000						
##	bwt	0.000	0.000					
##	wpre	0.065	0.000	0.000				
##	wgain	0.000	0.000	0.000	0.000			
##	education	0.000	0.000	0.002	0.004	0.187		
##	gestation	0.835	0.006	0.000	0.075	0.000	0.002	

Summary

- Collinearity of explanatory variables must be checked before deciding on a final multivariate linear regression model to ensure none of the explanatory variables are highly correlated.
- A correlation matrix can be created using the **corrplot** function.
- A matrix of p-values from correlation tests can be created using the **rcorr** function from the **Hmisc** package.

Linear Regression II

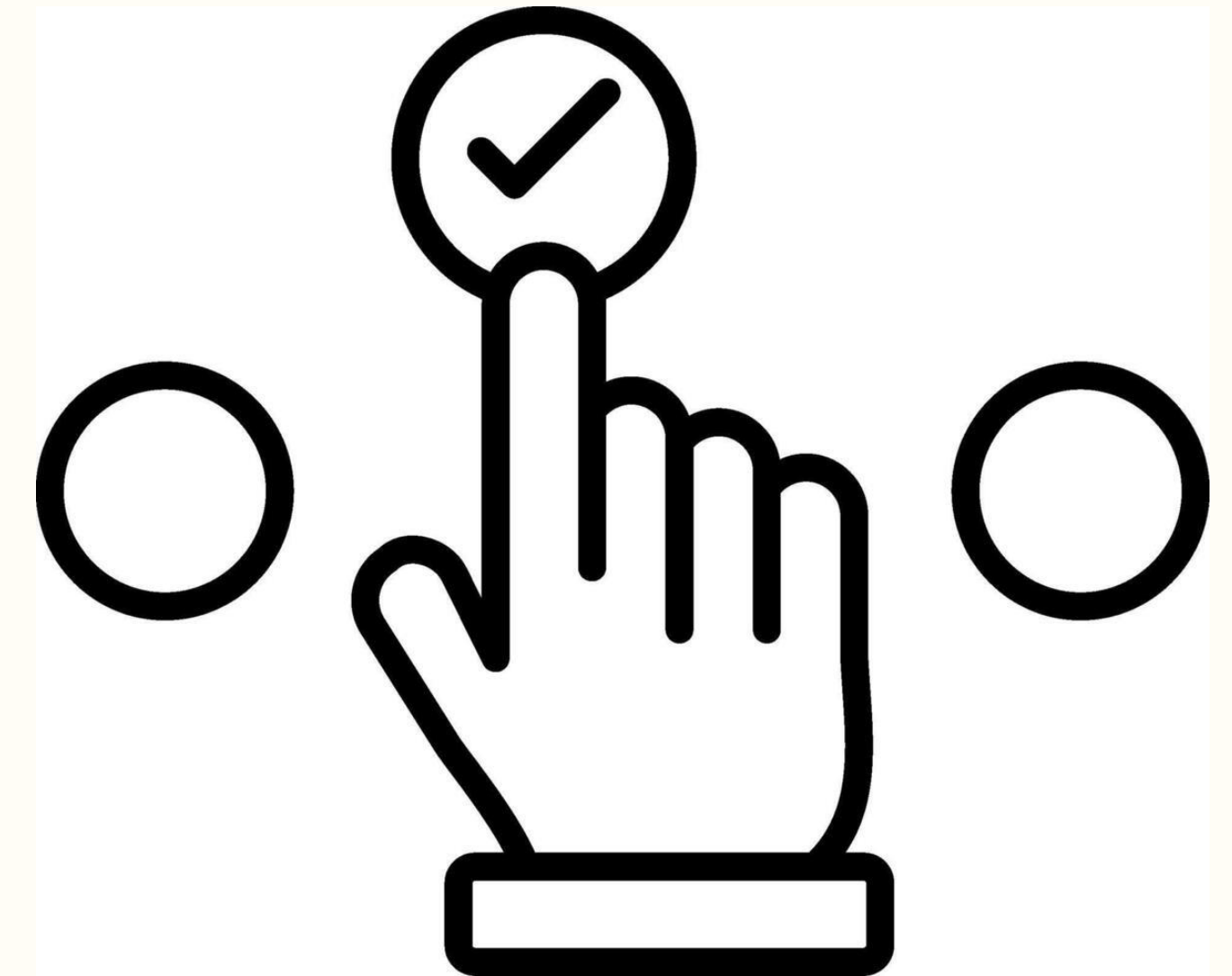
Model selection

Heather Mattie, PhD

Model selection

Model selection is the process of identifying the best set of covariates for a model.

- Depends on the type of model
- Prediction models should be parsimonious
- Causal models should control for known confounders and effect measure modification



Example: Prenatal care

```
data <- read.table("KingCounty2001_data.txt", header = TRUE)
head(data)
```

```
##   gender plural age   race parity married  bwt smokeN drinkN firststep welfare
## 1      F      1  31 hispanic      4      1 3118      0      0      1      0
## 2      M      1  23   white      0      1 3466      0      0      1      0
## 3      F      1  24   black      1      0 3147      0      0      1      0
## 4      M      1  26 hispanic      2      1 3969      0      0      1      0
## 5      M      1  34   white      0      0 3005      0      0      1      0
## 6      M      1  15   black      0      0 2920      0      0      1      0
##   smoker drinker wpre wgain education gestation
## 1      N      N  122   22          5          40
## 2      N      N  160   50         12          39
## 3      N      N  150   38         13          40
## 4      N      N  175   15         12          39
## 5      N      N  123   17         17          40
## 6      N      N  180   12          9          36
```

Prediction models

Prediction models are used to make future predictions of the outcome from a set of values from the explanatory variables.

- Only include explanatory variables that help predict or explain the variability in the outcome y
- This is called a **parsimonious** model
- Stepwise selection is often used for model selection
- Variables are added or removed until the best performing model is found

Forward selection

Forward selection starts with an empty model (containing only the intercept).

- Adds variables one by one, selecting the variable that provides the most significant improvement in model fit at each step
- Continues until adding no further variables significantly improves the model

```
# 1. Forward Selection
# Start with a null model (intercept only)
null_model <- lm(bwt ~ 1, data = data)

# Define the scope of the full model
full_scope <- formula(lm(bwt ~ firststep + gestation + gender +
                        age + welfare + smoker + drinker +
                        wpre + wgain + race, data = data))

forward_model <- step(null_model, direction = "forward", scope = full_scope)
```

Forward selection

- Output will show iterations
- Use the **summary** function to display the final model
- In this case the final model included **gestation**, **wpre**, **wgain**, **gender**, **smoker**, **race**, and **age**

```
print(summary(forward_model))
```

```
##
## Call:
## lm(formula = bwt ~ gestation + wpre + wgain + gender + smoker +
##      race + age, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2000.26  -276.27    -6.84   274.18  1521.24
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -2150.8297   157.6381  -13.644 < 2e-16 ***
## gestation     116.9456     3.7855   30.893 < 2e-16 ***
## wpre           3.1031     0.2664   11.650 < 2e-16 ***
## wgain          7.0537     0.6799   10.375 < 2e-16 ***
## genderM       134.3881    17.8121    7.545 6.31e-14 ***
## smokerY      -182.4192    35.6612   -5.115 3.37e-07 ***
## raceblack     -47.2143    40.9445   -1.153 0.248968
## racehispanic  130.9931    38.2908    3.421 0.000634 ***
## raceother      94.9887    83.7363    1.134 0.256745
## racewhite      90.3979    25.5975    3.532 0.000421 ***
## age           7.1036     1.5602    4.553 5.54e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 444.5 on 2489 degrees of freedom
## Multiple R-squared:  0.371, Adjusted R-squared:  0.3684
## F-statistic: 146.8 on 10 and 2489 DF, p-value: < 2.2e-16
```

Backward elimination

Backward elimination starts with a full model.

- Removes variables one by one, eliminating the variable that contributes the least to the model's fit
- Continues until removing no further variables significantly improve the model or all remaining variables are deemed significant.

```
# 2. Backward Elimination
# Start with a full model
full_model <- lm(bwt ~ firststep + gestation + gender +
                 age + welfare + smoker + drinker +
                 wpre + wgain + race, data = data)

backward_model <- step(full_model, direction = "backward")
```

Backward elimination

- Output will show iterations
- Use the **summary** function to display the final model
- In this case the final model included **gestation, wpre, wgain, gender, smoker, race, and age**

```
print(summary(backward_model))
```

```
##
## Call:
## lm(formula = bwt ~ gestation + gender + age + smoker + wpre +
##      wgain + race, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2000.26  -276.27   -6.84   274.18  1521.24
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -2150.8297   157.6381  -13.644 < 2e-16 ***
## gestation     116.9456     3.7855   30.893 < 2e-16 ***
## genderM       134.3881    17.8121    7.545 6.31e-14 ***
## age           7.1036      1.5602    4.553 5.54e-06 ***
## smokerY      -182.4192    35.6612   -5.115 3.37e-07 ***
## wpre          3.1031      0.2664   11.650 < 2e-16 ***
## wgain         7.0537      0.6799   10.375 < 2e-16 ***
## raceblack    -47.2143    40.9445   -1.153 0.248968
## racehispanic 130.9931    38.2908    3.421 0.000634 ***
## raceother    94.9887    83.7363    1.134 0.256745
## racewhite    90.3979    25.5975    3.532 0.000421 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 444.5 on 2489 degrees of freedom
## Multiple R-squared:  0.371, Adjusted R-squared:  0.3684
## F-statistic: 146.8 on 10 and 2489 DF, p-value: < 2.2e-16
```

Hybrid (bidirectional) selection

Hybrid (bidirectional) selection combines both forward and backward movements at each step.

- At each iteration, it considers adding a new variable (like forward selection) and removing an existing variable (like backward elimination)
- It selects the action (add or remove) that results in the greatest improvement in model fit
- This process continues until no further additions or removals improve the model

```
# 3. Hybrid Stepwise Selection  
# Can start from a null or full model, or any intermediate model  
hybrid_model <- step(null_model, direction = "both", scope = full_scope)
```

Hybrid (bidirectional) selection

- Output will show iterations
- Use the **summary** function to display the final model
- In this case the final model included **gestation**, **wpre**, **wgain**, **gender**, **smoker**, **race**, and **age**

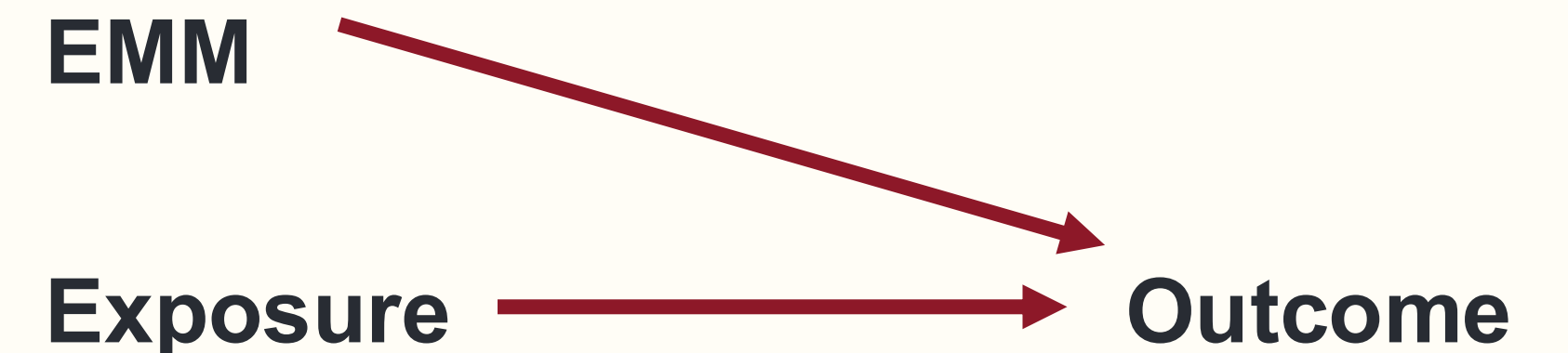
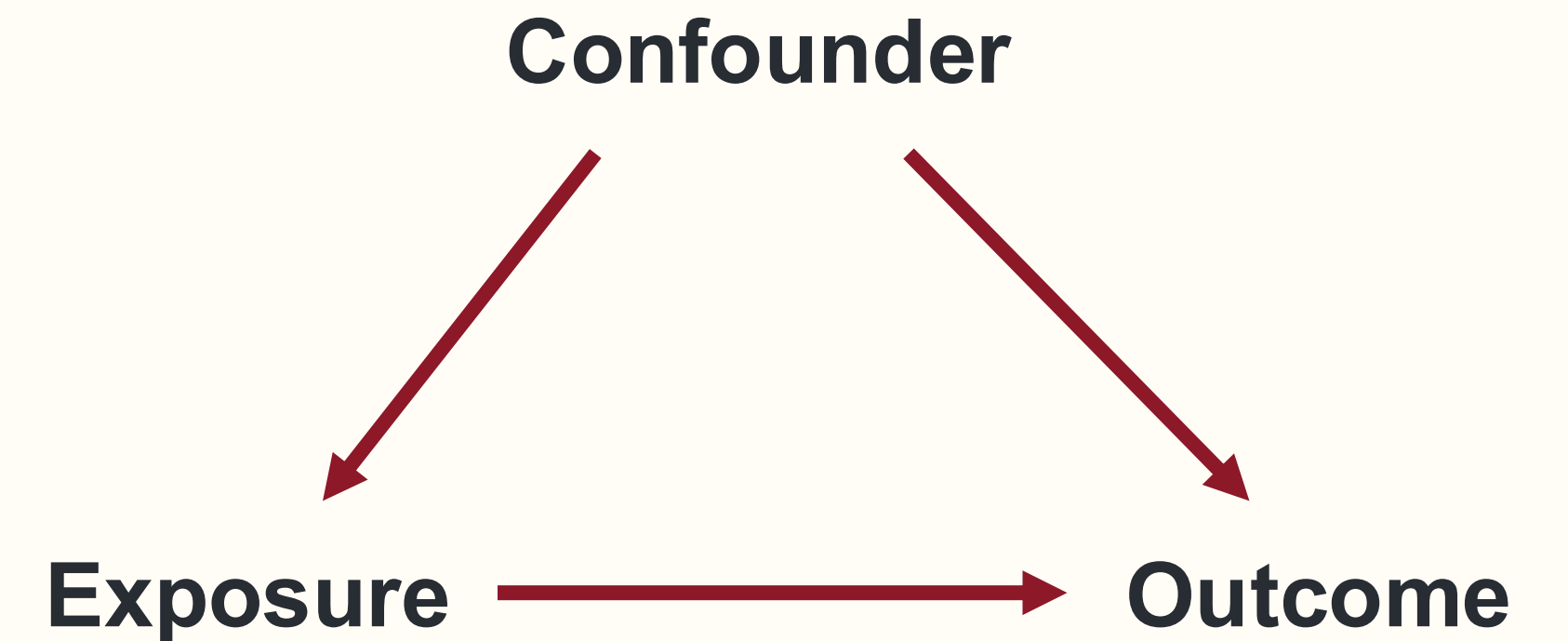
```
print(summary(hybrid_model))
```

```
##
## Call:
## lm(formula = bwt ~ gestation + wpre + wgain + gender + smoker +
##      race + age, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2000.26  -276.27   -6.84   274.18  1521.24
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -2150.8297   157.6381  -13.644 < 2e-16 ***
## gestation     116.9456     3.7855   30.893 < 2e-16 ***
## wpre           3.1031     0.2664   11.650 < 2e-16 ***
## wgain         7.0537     0.6799   10.375 < 2e-16 ***
## genderM       134.3881    17.8121    7.545 6.31e-14 ***
## smokerY      -182.4192    35.6612   -5.115 3.37e-07 ***
## raceblack     -47.2143    40.9445   -1.153 0.248968
## racehispanic  130.9931    38.2908    3.421 0.000634 ***
## raceother      94.9887    83.7363    1.134 0.256745
## racewhite      90.3979    25.5975    3.532 0.000421 ***
## age           7.1036     1.5602    4.553 5.54e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 444.5 on 2489 degrees of freedom
## Multiple R-squared:  0.371, Adjusted R-squared:  0.3684
## F-statistic: 146.8 on 10 and 2489 DF, p-value: < 2.2e-16
```

Causal models

Causal models estimate the effect of an exposure or intervention on an outcome, while accounting for confounding factors.

- Control for confounding
- Test and account for effect measure modification



Causal model

- Known confounders are retained in the model, whether statistically significant or not.
- Effect measure modifiers are retained only if the interaction term is significant.

```
m <- lm(bwt ~ firststep + gestation + age + welfare +  
        smoker + drinker + wpre + wgain + race,  
        data = data)  
summary(m)
```

Causal model

- Keep **firststep** because it is the exposure of interest
- Keep all other covariates because they have been shown to be confounders in previous literature

```
##
## Call:
## lm(formula = bwt ~ firststep + gestation + age + welfare + smoker +
##      drinker + wpre + wgain + race, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2059.49  -286.96    8.57   275.98  1582.21
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -2036.6494   159.6849  -12.754 < 2e-16 ***
## firststep      -13.0216    26.2990   -0.495 0.620548
## gestation     115.7529     3.8333   30.197 < 2e-16 ***
## age           6.9592     1.6199    4.296 1.80e-05 ***
## welfare      -97.4142    72.0268   -1.352 0.176347
## smokerY      -174.2710    36.4732   -4.778 1.87e-06 ***
## drinkerY      -69.8747    84.4109   -0.828 0.407867
## wpre          3.1275     0.2701   11.578 < 2e-16 ***
## wgain         7.1616     0.6877   10.413 < 2e-16 ***
## raceblack     -40.4257    41.8529   -0.966 0.334188
## racehispanic  133.9967    39.0928    3.428 0.000619 ***
## raceother     117.0642    84.8951    1.379 0.168041
## racewhite      92.9300    25.9126    3.586 0.000342 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 449.5 on 2487 degrees of freedom
## Multiple R-squared:  0.3573, Adjusted R-squared:  0.3542
## F-statistic: 115.2 on 12 and 2487 DF, p-value: < 2.2e-16
```

Summary

- Model selection depends on the purpose of the model.
- When building a prediction model, the goal is to build a parsimonious model that includes explanatory variables that help predict the outcome.
- When building a causal model, the goal is to build a model that focuses on the relationship between one exposure and outcome, and that adjusts for confounding and effect measure modification.